OXFORD

human reproduction update

Human Reproduction Update, 2023, **29(6)**, 741–772
https://doi.org/10.1093/humupd/dmad015
Advance Access Publication Date: June 23, 2023

PCOS during the menopausal transition and after menopause: a systematic review and meta-analysis

Mercedes Millán-de-Meer[1,4,†], Manuel Luque-Ramírez [1,2,3,†], Lía Nattero-Chávez [2,3], and Héctor F. Escobar-Morreale [1,2,3,*]

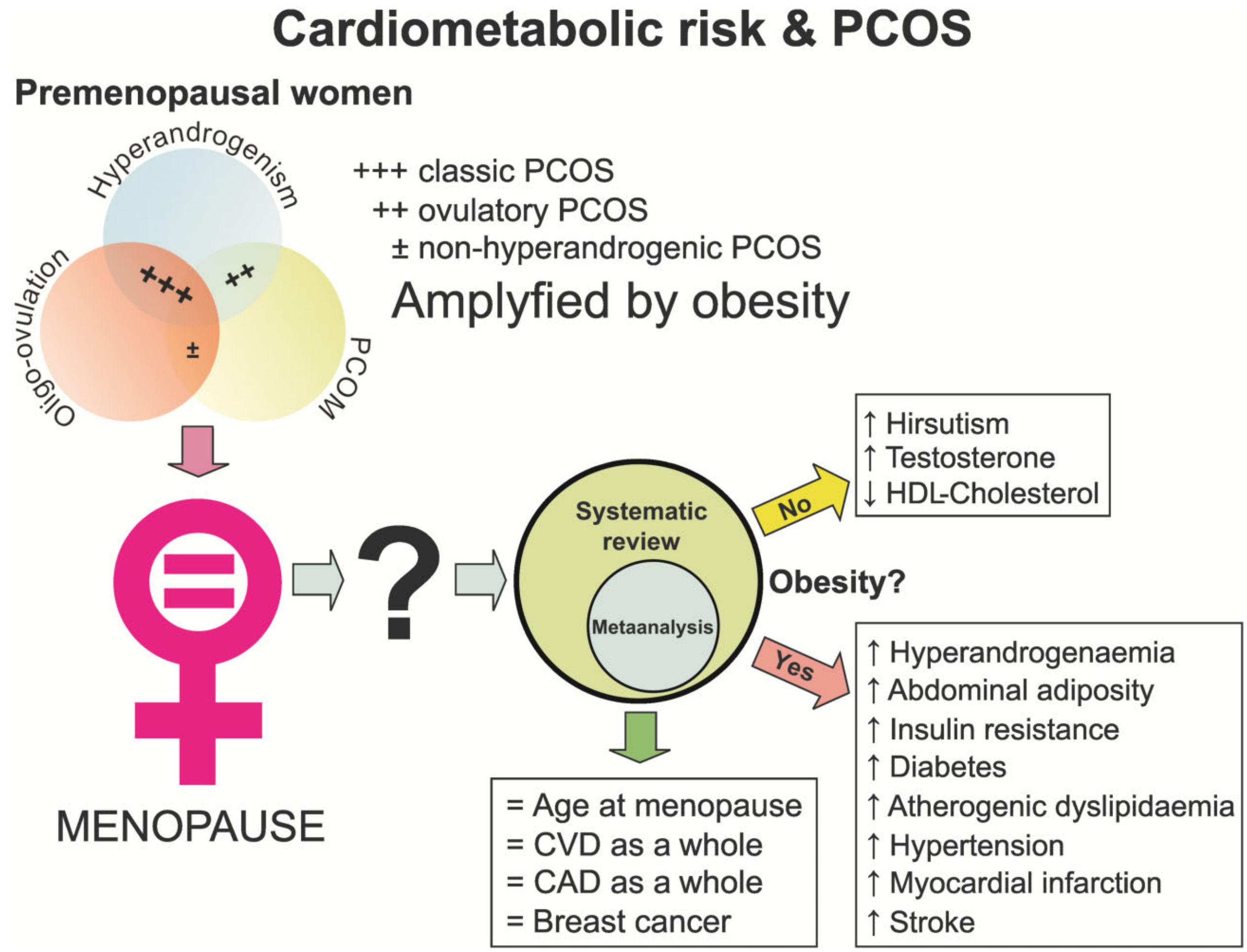
[1]Universidad de Alcalá, Madrid, Spain
[2]Department of Endocrinology and Nutrition, Hospital Universitario Ramón y Cajal, Madrid, Spain
[3]Grupo de Diabetes, Obesidad y Reproducción Humana, Universidad de Alcalá & Instituto Ramón y Cajal de Investigación Sanitaria (IRYCIS) & Centro de Investigación Biomédica en Red de Diabetes y Enfermedades Metabólicas Asociadas (CIBERDEM), Madrid, Spain
[4]Present address: Department of Anaesthesia and Resuscitation. Hospital Universitario La Paz, Madrid, Spain

*Correspondence address. Department of Endocrinology & Nutrition, Hospital Universitario Ramón y Cajal, Ctra. de Colmenar Viejo km 9.1, 28034 Madrid, Spain. E-mail: hectorfrancisco.escobar@salud.madrid.org https://orcid.org/0000-0002-6890-1644
†These authors contributed equally to this work.

TABLE OF CONTENTS

Received: February 21, 2023. **Revised:** May 17, 2023. **Editorial decision:** May 28, 2023.

GRAPHICAL ABSTRACT



Systematic review and meta-analysis suggest that only hyperandrogenism persists after menopause in women with PCOS, whilst most cardiometabolic comorbidities relate to the coexistence of obesity.

ABSTRACT

BACKGROUND: Current knowledge about the consequences of PCOS during the late reproductive years and after menopause is limited.

OBJECTIVE AND RATIONALE: We performed a systematic review and meta-analysis of data on the pathophysiology, clinical manifestations, diagnosis, prognosis, and treatment of women ≥45 years of age—peri- or postmenopausal—with PCOS.

SEARCH METHODS: Studies published up to 15 April 2023, identified by Entrez-PubMed, EMBASE, and Scopus online facilities, were considered. We included cross-sectional or prospective studies that reported data from peri- or postmenopausal patients with PCOS and control women with a mean age ≥45 years. Three independent researchers performed data extraction. Meta-analyses of quantitative data used random-effects models because of the heterogeneity derived from differences in study design and criteria used to define PCOS, among other confounding factors. Sensitivity analyses restricted the meta-analyses to population-based studies, to studies including only patients diagnosed using the most widely accepted definitions of PCOS, only menopausal women or only women not submitted to ovarian surgery, and studies in which patients and controls presented with similar indexes of weight excess. Quality of evidence was assessed using the GRADE system.

OUTCOMES: The initial search identified 1400 articles, and another six were included from the reference lists of included articles; 476 duplicates were deleted. We excluded 868 articles for different reasons, leaving 37 valid studies for the qualitative synthesis, of which 28 studies—published in 41 articles—were considered for the quantitative synthesis and meta-analyses. Another nine studies were included only in the qualitative analyses. Compared with controls, peri- and postmenopausal patients with PCOS presented increased circulating total testosterone (standardized mean difference, SMD 0.78 (0.35, 1.22)), free androgen index (SMD 1.29 (0.89, 1.68)), and androstenedione (SMD 0.58 (0.23, 0.94)), whereas their sex hormone-binding globulin was reduced (SMD −0.60 (−0.76, −0.44)). Women with PCOS showed increased BMI (SMD 0.57 (0.32, 0.75)), waist circumference (SMD 0.64 (0.42, 0.86)), and waist-to-hip ratio (SMD 0.38 (0.14, 0.61)) together with increased homeostasis model assessment of insulin resistance (SMD 0.56 (0.27, 0.84)), fasting insulin (SMD 0.61 (0.38, 0.83)), fasting glucose (SMD 0.48 (0.29, 0.68)), and odds ratios (OR, 95% CI) for diabetes (OR 3.01 (1.91, 4.73)) compared to controls. Women with PCOS versus controls showed decreased HDL concentrations (SMD −0.32 (−0.46, −0.19)) and increased triglycerides (SMD 0.31 (0.16, 0.46)), even though total cholesterol and LDL concentrations, as well as the OR for dyslipidaemia, were similar to those of controls. The OR for having hypertension was increased in women with PCOS compared with controls (OR 1.79 (1.36, 2.36)). Albeit myocardial infarction (OR 2.51 (1.08, 5.81)) and stroke (OR 1.75 (1.03, 2.99)) were more prevalent in women with PCOS than controls, the ORs for cardiovascular disease as a whole, coronary artery disease as a whole, breast cancer and age at menopause, were similar in patients and controls. When restricting meta-analysis to studies in which women with PCOS and controls had a similar mean BMI, the only difference that retained statistical significance was a decrease in HDL-cholesterol concentration in the former and, in the two studies in which postmenopausal women with PCOS and controls had similar BMI, patients presented with increased serum androgen concentrations, suggesting that hyperandrogenism persists after menopause, regardless of obesity.

WIDER IMPLICATIONS: Hyperandrogenism appeared to persist during the late-reproductive years and after menopause in women with PCOS. Most cardiometabolic comorbidities were driven by the frequent coexistence of weight excess and PCOS, highlighting the importance of targeting obesity in this population. However, the significant heterogeneity among included studies, and the overall low quality of the evidence gathered here, precludes reaching definite conclusions on the issue. Hence, guidelines derived from adequately powered prospective studies are definitely needed for appropriate management of these women.

Keywords: androgen excess / cancer / cardiovascular disease / diabetes / dyslipidaemia / hirsutism / hyperandrogenism / hypertension / prevalence / virilisation

Introduction

PCOS is possibly the most common endocrine disorder in women of reproductive age, with a worldwide prevalence of 6–15% in premenopausal women from the general population (Asuncion *et al.*, 2000; Yildiz *et al.*, 2012).

A mainly androgen excess disorder, PCOS results from a primary abnormality in ovarian and adrenal steroidogenesis (Wickenheisser *et al.*, 2006; Escobar-Morreale, 2018). Androgen excess may facilitate a predominantly visceral deposition of body fat and adipose tissue dysfunction, leading to insulin resistance and compensatory hyperinsulinism (Escobar-Morreale and San Millan, 2007). The hyperinsulinism increases androgen secretion by co-stimulating, beside gonadotrophins and adrenocorticotrophin, the ovaries and the adrenals (Diamanti-Kandarakis and Dunaif, 2012), closing a vicious circle whereby androgen excess favouring the abdominal deposition of fat further facilitates androgen secretion by these glands in women with PCOS (Escobar-Morreale and San Millan, 2007; Escobar-Morreale, 2018).

The common association of PCOS with weight excess, obesity, and insulin resistance predisposes premenopausal women with PCOS towards glucose intolerance and type 2 diabetes mellitus (Ortiz-Flores *et al.*, 2019), hypertension (Luque-Ramirez *et al.*, 2014), atherogenic dyslipidaemia (Wild *et al.*, 2011; Zhu *et al.*, 2019), cerebrovascular disease (Wekker *et al.*, 2020), and an increased cardiovascular risk (Wild *et al.*, 2010), even though the association of PCOS with actual cardiovascular events remains unclear (Legro, 2003; Teede *et al.*, 2018; Dokras, 2022). Aside from subfertility, chronic oligo-amenorrhoea predisposes women with PCOS to endometrial carcinoma, even though no association has been found with breast and ovarian cancer (Barry *et al.*, 2014). On the contrary, indirect data derived from their increased anti-Müllerian hormone concentrations may suggest that menopause might occur later in women with PCOS compared with the general population (Minooee *et al.*, 2018).

Of note is that all the evidence supporting the current diagnosis and management of PCOS is derived from data obtained from women during their reproductive years. In contrast, current data regarding PCOS during the menopausal transition and after menopause are scarce, particularly with regards to the clinical manifestations of the syndrome and its cardiometabolic comorbidities.

We aimed to conduct a systematic review, followed by meta-analysis of quantitative data whenever possible, of the pathophysiology, clinical manifestations, diagnosis, prognosis, and treatment of PCOS in peri- or postmenopausal women.

Methods

The review protocol was registered in the Open Science Framework registry for systematic review protocols (https://osf.io/g5pv4/, last accessed, 30 January 2023), and followed the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) (Page *et al.*, 2021).

PICO concept

Patients: peri- and postmenopausal women with PCOS; Interventions: observation; Comparison: peri- and postmenopausal women without PCOS; Outcomes: any data regarding the pathophysiology, clinical manifestations, diagnosis, prognosis, and treatment of PCOS.

Quantitative outcomes submitted to meta-analysis

Age at menopause (year-old); BMI (kg/m^2); waist circumference (cm) and waist-to-hip ratio (cm/cm); total testosterone (TT, nmol/l); free androgen index (FAI, no units); androstenedione (nmol/l); dehydroepiandrosterone-sulphate (DHEAS, μmol/l); sex hormone-binding globulin (SHBG, nmol/l); homeostasis model assessment of insulin resistance (HOMA-IR, no units); fasting insulin (pmol/l); fasting glucose (mmol/l); AUC of the oral glucose tolerance test of insulin (pmol/l/120 min) and glucose (mmol/l/120 min); composite insulin sensitivity index (no units); odds ratio (OR) for diabetes; total cholesterol (mmol/l); high-density lipoprotein cholesterol (HDL-cholesterol, mmol/l); low-density lipoprotein cholesterol (LDL-cholesterol, mmol/l); triglycerides (TG, mmol/l); OR for dyslipidaemia; systolic (SBP, mmHg) and diastolic (DBP, mmHg) blood pressure; OR for hypertension; carotid intima-media thickness (cIMT, mm); OR for cardiovascular disease (CVD); OR for coronary artery disease; OR for myocardial infarction; OR for stroke; and OR for breast cancer.

Data sources and searches

We used the online facilities of Entrez-PubMed®, EMBASE®, and Scopus®. The electronic search strategy, including dates, terms, filters, and limits applied, are described in detail in Supplementary Table S1. In addition, the reference lists of included articles were scrutinized to detect relevant studies.

Study selection

We included cross-sectional or prospective studies that contained data on the pathophysiology, clinical presentation, diagnosis, treatment, and prognosis of PCOS, as defined by currently accepted definitions of the syndrome such as the 1990 National Institutes of Health (NIH) criteria (Zawadzki and Dunaif, 1992), the 2003 ESHRE and American Society for Reproductive Medicine (ASRM) criteria (The Rotterdam ESHRE/ASRM-Sponsored PCOS Consensus Workshop Group, 2004), or the 2006 Androgen Excess & Polycystic Ovary Syndrome Society (AE-PCOS) criteria (Azziz *et al.*, 2006) or by histopathology, or studies describing PCOS phenotypes and consequences during menopause. No restriction by type of study was applied; however, we excluded review articles, case reports, and articles not submitted to a strict peer-review process such as congress abstracts.

For the present analysis, we included only studies in which the mean age of the group of patients was 45 years or older, considering a 47.5 years median age of inception of perimenopause (McKinlay *et al.*, 1992). This means that some of the women included in the studies analyzed here could be younger than 45 years, but were, nevertheless, in their late reproductive years. Studies reporting quantitative variables in women with PCOS and appropriate controls were submitted to meta-analysis.

In the first stage, screening of titles and abstracts was followed by the retrieval and screening of full-text articles using these inclusion criteria. Three reviewers (M.M.-d.M., M.L.-R., and L.N.-C.) independently assessed the references, and disagreements were resolved by discussion to reach consensus. If several publications were reported on the same study, the publication that provided the larger amount of data about the same outcome variable was included in the meta-analysis. The searches were re-run just before the final analyses on 15 April 2023 and further studies retrieved for inclusion.

Data synthesis and analysis

Only outcomes for which valid data were available in at least three studies were submitted to meta-analysis. For continuous variables, we calculated the weighed mean difference (MD) and the standardized mean difference (SMD) between groups, and their 95% CI. SMD effects were calculated back to expected change in percentile rank of every outcome for average women with PCOS compared with average controls using Cohen's *U3* index. Data were introduced as mean and SD; when other statistics were reported in the original studies—such as medians, range, or quartiles—we estimated the mean and SD according to Wan *et al.* (2014) and to Taylor equations (https://www.cebm.ox.ac.uk/resources/data-extraction-tips-meta-analysis/missing-mean, last accessed 30 January 2023). In the few cases when the mean could not be calculated, we introduced the medians instead of the means. Heterogeneity and inconsistency across studies were assessed by χ^2 test and Higgins's I^2 statistics. An $I^2 > 50\%$ was considered as a substantial heterogeneity. We applied a random effects model to weight studies; the SD of underlying effects across studies was estimated as Tau^2. For dichotomous data, results for each study were expressed as odds ratios (OR) with 95% CI for patients with PCOS presenting the outcome compared with control women, and combined for meta-analysis using the Mantel–Haenszel random effects method.

For sensitivity analyses, we planned separate meta-analyses of studies using strict definitions for the diagnosis of PCOS, population-based studies, studies reporting only women with established menopause, women without a history of ovarian surgery of any kind, and studies in which patients and controls had similar indexes of weight excess.

We used RevMan 5.4 (https://training.cochrane.org/online-learning/core-software/revman/revman-5-download, last accessed 30 January 2023) for meta-analyses. When 10 or more studies were analyzed, publication bias was assessed by funnel plots (Sterne *et al.*, 2005), as shown in Supplementary Fig. S1. All *P*-values were two-sided, and $\alpha = 0.05$ was set as level of statistical significance.

Quality assessment

Two authors (M.L.-R. and L.N.-C.) independently assessed the risk of bias of individual studies included in the diverse meta-analyses using the 'ROBINS-E' tool (Risk Of Bias In Non-randomized Studies—of Exposures, https://www.riskofbias.info/welcome/robins-e-tool, last accessed 30 January 2023). When needed, we resolved disagreements over the risk of bias by discussion to reach consensus or arbitration by H.F.E.-M. We summarized these results by using 'robvis' visualization tool (https://www.riskofbias.info/welcome/robvis-visualization-tool, last accessed 30 January 2023). We displayed colour-blind friendly 'traffic light' plots of the domain-levels judgement for each individual result, and weighted bar plots of the distribution of risk-of-bias judgements within each bias domain (McGuinness and Higgins, 2021).

Then, in order to address the quality of evidence derived from our meta-analyses for primary outcomes separately, after assessing study limitations (risk of bias) as outlined above, we used the Grading of Recommendations, Assessments, Development, and Evaluation (GRADE) system (https://gradepro.org, last accessed 30 January 2023) with the requirements specified in the GRADE Handbook for grading the quality of evidence for systematic reviews (http://gdt.guidelinedevelopment.org/app/handbook/handbook.html, last accessed 30 January 2023). Results were summarized using the GRADEpro GDT free online application (https://gradepro.org, last accessed 30 January 2023).

Results

Outcome of the search

The initial search identified 1400 articles, another six were included from the reference lists of included articles, and 476 duplicates were deleted. We excluded 868 articles for a variety of reasons, leaving 37 valid studies for the qualitative synthesis, of which 28 studies—published in 41 articles—were considered for the quantitative synthesis and meta-analyses. Another nine studies were included only in the qualitative analyses (Fig. 1 and Table 1).

Risk of bias of studies included in the meta-analyses

A plot of the domain-levels judgement for each individual study and of the distribution of risk-of-bias judgements within each bias domain is shown in Supplementary Fig. S2. Overall risk of bias was high or very high for all studies. While bias arising from measurement of the outcome and bias derived from reporting of selected results were considered low, the main reasons for high-very high risk of bias were: bias owing to post-exposure interventions since a woman with PCOS may be more likely to receive some kind of diagnostic or therapeutic intervention throughout her lifetime than a non-hyperandrogenic women, affecting studied outcomes; bias in selection of participants into different studies since several studies based PCOS diagnosis on participant characteristics likely to be influenced by the outcome or a cause of the outcome (e.g. insulin resistance indexes or postmenopausal hyperandrogenaemia); bias arising from measurement of the exposure (i.e. non-standard definitions of PCOS); and bias caused by under-adjusted or uncontrolled confounding factors when such a bias was sufficient high to threaten conclusions about the effect of PCOS status on the outcomes.

Quality of evidence

The quality of evidence ranged from very low to moderate, according to the GRADE system. Table 2 shows a summary of the quality of evidence grades for selected outcomes. Quality of evidence was moderate for waist circumference and circulating HDL-cholesterol associations, low for BMI, FAI, SHBG, myocardial infarction, and breast cancer, and very low for the remaining outcomes in the meta-analyses.

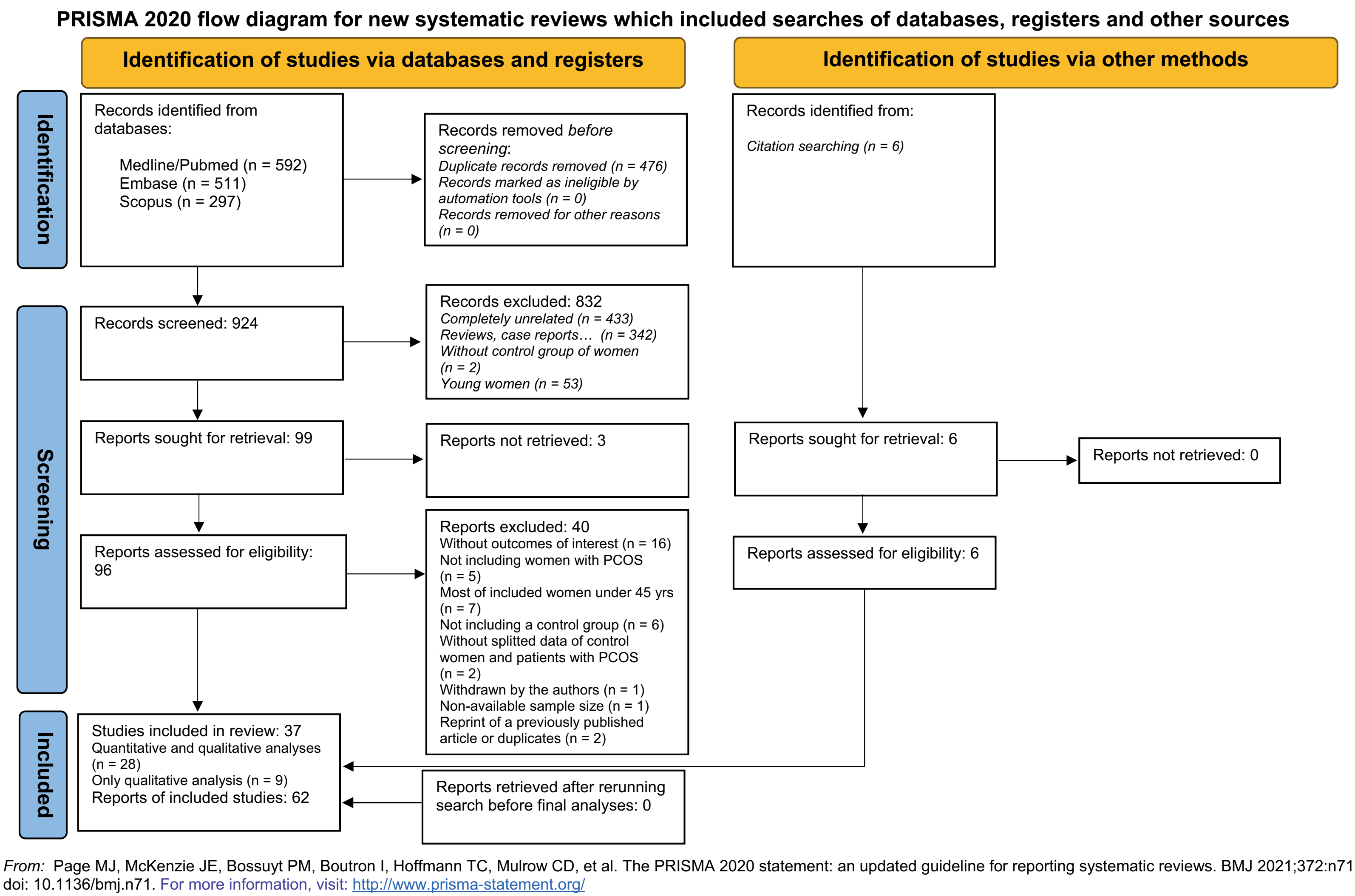


Figure 1. PRISMA 2020 flowchart for the selection of studies in a systematic review of PCOS during the menopausal transition and after menopause.

Table 1. Studies and reports included in the meta-analyses or in the descriptive review of PCOS during the menopausal transition and after menopause.

Study	Report (Author)	Year	Region	Study design/ Recruitment/Setting	Mean age of PCOS population (years, mean±SD or range)	BMI matched	PCOS phenotype
INCLUDED IN META-ANALYSIS							
Alsamarai *et al.*		2009	USA	Case control and longitudinal cohort/ Referral	46 ± 5	No	NIH 1990 (classic PCOS, fulfilling Rotterdam criteria)
The APOS study	Elting *et al.*	2001	The Netherlands	Longitudinal cohort/ Referral	45–54	No	Oligo- or amenorrhoea and an increased LH concentration, determined at least 2 weeks after and 3 weeks before a menstrual bleeding, in the presence of a normal FSH concentration
Armeni *et al.*		2013	Greece	Case–control/Referral	56 ± 8*	No	Women fulfilling at least 3 of the following criteria: (i) ovulatory dysfunction during their reproductive years, (ii) insulin resistance or elevated plasma glucose, (iii) central obesity, or (iv) clinical or biochemical hyperandrogenism
The Cancer and Steroid Hormone Study	Gammon & Thompson	1991	USA	Case–control/ Population	Only postmenopausal women	No	History of physician-diagnosed polycystic ovaries ascertained through direct questioning during a structured interview administered by trained personnel
The Cardiovascular Health and Risk Measurement Study	Snyder *et al.*	2014	USA	Case–control/Referral	47 ± 5 Menopausal status: 37%	No	NIH 1990 criteria retrospectively allocated (classic PCOS, fulfilling Rotterdam criteria)
Chan *et al.*		2013	Australia	Case–control/Referral	55–69	No	Rotterdam criteria
Cibula *et al.*		2000	Czech Republic	Case–control/Referral	52 ± 5	Yes	NIH 1990 criteria (classic PCOS)+ovarian histopathology typical of PCOS at wedge resection
Echiburú *et al.*		2016	Chile	Case–control/Referral	46 ± 5	Yes	NIH 1990 (classic PCOS, fulfilling Rotterdam criteria)
The ELSA-Brasil study	Gabrielli *et al.*	2015	Brasil	Longitudinal cohort/ Population	55 ± 9 Menopausal status: 69%	No	Menopausal women with 2 of the following 3 features: (i) clinical or biochemical hyperandrogenism, (ii) menstrual dysfunction, and (iii) insulin resistance (HOMA-IR>2.2)
The Gothenburg's PCOS cohort I	Dahlgren *et al.*	1992	Sweden	Longitudinal cohort/ Referral	50 ± 5 Menopausal status: 30%	Yes	Ovarian histopathology typical of PCOS at wedge resection
		1994			53 ± 5 Menopausal status: 29%		Ovarian histopathology typical of PCOS at wedge resection
	Schmidt *et al.*	2011 2011 2017			70 ± 5*		Ovarian histopathology typical of PCOS at wedge resection (fulfilling Rotterdam criteria)
	Forslund *et al.*	2021 2022			81 ± 5*		Ovarian histopathology typical of PCOS at wedge resection (fulfilling Rotterdam criteria)
The Gothenburg's PCOS cohort II	Forslund *et al.*	2019 2020 2022		Longitudinal cohort/ Referral	52 ± 5 Menopausal status: 37%	No	NIH 1990 criteria (classic PCOS, fulfilling Rotterdam criteria)

(continued)

Table 1. (continued)

Study	Report (Author)	Year	Region	Study design/ Recruitment/Setting	Mean age of PCOS population (years, mean±SD or range)	BMI matched	PCOS phenotype
Guoquing *et al.*		2016	China	Case–control/Referral	55 ± 3*	No	NIH 1990 criteria retrospectively allocated (classic PCOS, fulfilling Rotterdam criteria)
The Iowa Women's Health Study	Anderson *et al.*	1997	USA	Longitudinal cohort/ Population	55–69	No	Self-reported
The London School of Hygiene & Tropical Medicine cohort	Wild *et al.*	2000	UK	Longitudinal cohort/ Referral	>45	No	Women identified with a PCOS diagnosis from histopathology records, operating theatre records of women who had undergone culdoscopy, wedge resection or biopsy of the ovary, admission and discharge records and diagnostic indexes, and presenting with history of oligomenorrhoea, acne, infertility or hirsutism
The Long Island Breast Cancer Study Project	Kim *et al.*	2016	USA	Case–control/ Population	54 ± 12 Menopausal status: 56%	Yes	Self-reported Rotterdam criteria by a structured questionnaire
Margolin *et al.*		2005	Israel	Case–control/ Population	56 ± 14*	No	Hyperandrogenemia or hirsutism plus past history of irregular menses
Markopoulos *et al.*		2013	Greece	Case–control/Referral	55 ± 5*	Yes	NIH 1990 criteria retrospectively allocated (classic PCOS, fulfilling Rotterdam criteria)
The POPCORn study	Meun *et al.*	2020	The Netherlands	Longitudinal cohort/ Population	51 ± 6 Menopausal status: 13%	No	Rotterdam criteria
The Northern Finland Birth Cohort 1996	Ollila *et al.*	2016 2017 2019	Finland	Longitudinal cohort/ Population	46 ± 0	No	PCOS diagnosed by a validated self-reported questionnaire
The Nordic Multicenter Collaboration study	Pinola *et al.*	2015	Finland Norway Sweden	Case–control/Referral	46 ± 5	No	Rotterdam criteria or hyperandrogenemia plus past history of oligomenorrhea in peri- and postmenopausal participants
Parazzini *et al.*		1997	Italy	Case–control/ Population	Median age 55	No	Self-reported Stein-Leventhal syndrome by a structured questionnaire
Persson *et al.*		2023	Sweden	Longitudinal cohort/ Referral	Median age 51	No	NIH-2012 or Rotterdam criteria present at 45 years
The Pittsburg University's PCOS cohort	Talbott *et al.*	2000 2008	USA	Longitudinal cohort/ Referral	>45 47 ± 6 Premenopausal status: 23%	No	Clinical diagnosis of PCOS was made if there was: (i) a history of chronic anovulation in association with (ii) hirsutism or biochemical evidence of an elevated total testosterone concentration (>2.0 nmol/L) or with (3) a ratio of LH/FSH >2
Puurunen *et al.*		2009 2011	Finland	Case–control/Referral	48 ± 5 52 ± 6		Rotterdam criteria in premenopausal women and oligomenorrhea or menstrual irregularity combined with hyperandrogenism reported previously in their medical records in postmenopausal ones
The Rancho Bernardo study	Krentz *et al.*	2007	USA	Longitudinal cohort/ Population	72 ± 9*	No	Presence of ≥3 features: (i) recalled history of irregular menses, (ii) clinical premenopausal hyperandrogenism or postmenopausal hyperandrogenenemia, (iii) history of infertility or miscarriage, (iv) central obesity, or (v) biochemical insulin resistance
The Rotterdam study	Meun *et al.*	2018	The Netherlands	Longitudinal cohort/ Population	70 ± 9*	No	Hyperandrogenemia plus past history of irregular menses

(continued)

Table 1. (continued)

Study	Report (Author)	Year	Region	Study design/ Recruitment/Setting	Mean age of PCOS population (years, mean±SD or range)	BMI matched	PCOS phenotype
The SWAN study	Polotsky *et al.*	2012 2014 2014	USA	Longitudinal cohort/Population	46 ± 3 Premenopausal status: 53% 46 ± 3 Premenopausal status: 34%	No	Hyperandrogenemia plus past history of irregular menses
The WISE study	Merz *et al.*	2016	USA	Longitudinal cohort/ Referral	63 ± 12*	Yes	Hyperandrogenemia plus past history of irregular menses
INCLUDED ONLY IN DESCRIPTIVE REVIEW							
Birdsall & Farquhar		1996	New Zealand	Case–control/Referral	>50	No	Polycystic ovarian morphology
Cheang *et al.*		2008	USA	Case–control/Referral	57 ± 8 Menopausal status >80%	No	Menstrual irregularity or subfertility history plus concurrent presence of clinical signs of hyperandrogenemia in mothers of women with PCOS
Deniz & Kehribar		2021	Turkey	Case–control/Referral	>50*	No	NIH 2012 criteria (fulfilling Rotterdam criteria)
Elhasan *et al.*		2018	UK	Case–control/Referral	>50*	No	Rotterdam criteria
Gothenburg's PCOS cohort I	Schmidt *et al.*	2012	Sweden	Longitudinal cohort/ Referral	69 ± 5*	Yes	Ovarian histopathology typical of PCOS at wedge resection (fulfilling Rotterdam criteria)
Iatrakis *et al.*		2006	Greece	Case–control/Referral	46	No	Confirmed diagnosis of PCOS (non-specified criteria)
The Karma study	Li *et al.*	2016	Sweden	Longitudinal cohort/ Population	Menopausal women	No	Self-reported
Markopoulos *et al.*		2011	Greece	Case–control/Referral	55 ± 4*	Yes	NIH 1990 criteria retrospectively allocated (fulfilling Rotterdam criteria)
The Midlife Women's Health Study	Yin *et al.*	2018	USA	Longitudinal cohort/ Population	48	Yes	Rotterdam criteria
The Northern Finland Birth Cohort	Karjula *et al.* Pölönen *et al.* Kujanpää *et al.*	2017 2020 2022 2022	Finland	Longitudinal cohort/ Population	46 ± 0	No	PCOS diagnosed by a validated self-reported questionnaire
The Pittsburg University's PCOS cohort	Talbott *et al.* Winters *et al.*	1998 2004 2004 2000	USA	Longitudinal cohort/ Referral	>45 48 ± 5 49 ± 4 >47	Yes	History of chronic anovulation in association with either clinical evidence of androgen excess (hirsutism) or if total testosterone concentration was >2 nmol/L or the LH/FSH ratio was greater than 2 by revision of clinical records
The Rancho Bernardo study	Krentz *et al.*	2009 2012	USA	Longitudinal cohort/ Population	72 ± 9*	No	Presence of ≥3 features: (i) recalled history of irregular menses, (ii) clinical premenopausal hyperandrogenism or postmenopausal hyperandrogenenemia, (iii) history of infertility or miscarriage, (iv) central obesity, or (v) biochemical insulin resistance
Zucchetto *et al.*		2009	Italy	Case–control/Referral	Median age 60	No	Self-reported PCOS collected by centrally trained interviewers using a structured questionnaire

* All PCOS participants were postmenopausal women.

Table 2. Quality of evidence for the outcomes of meta-analyses according to the GRADE Working Group.

Outcome	Risk of bias	Inconsistency of results	Indirectness of evidence	Imprecision	Other considerations	Quality of evidence
CLINICAL MANIFESTATIONS						
Age at menopause	Very serious	Very serious	Not serious	Not serious	None	⊕○○○ Very low
BMI	Very serious	Not serious	Serious	Not serious	Strong association Plausible confounding would suggest spurious association	⊕⊕○○ Low
Waist circumference	Very serious	Not serious	Not serious	Not serious	Strong association Plausible confounding would suggest spurious association	⊕⊕⊕○ Moderate
Waist-to-hip ratio	Very serious	Serious	Not serious	Serious	Strong association Plausible confounding would suggest spurious association	⊕○○○ Very low
CIRCULATING ANDROGEN PROFILES						
Total testosterone	Very serious	Serious	Not serious	Serious	Strong association All plausible residual confounding would reduce demonstrated effect	⊕○○○ Very low
Free androgen index	Very serious	Not serious	Not serious	Serious	Strong association All plausible residual confounding would reduce demonstrated effect	⊕⊕○○ Low
Androstenedione	Very serious	Serious	Not serious	Serious	Strong association All plausible residual confounding would reduce demonstrated effect	⊕○○○ Very low
Dehydroepiandrosterone-sulphate	Very serious	Serious	Not serious	Serious	All plausible residual confounding would reduce demonstrated effect	⊕○○○ Very low
Sex hormone-binding globulin	Very serious	Serious	Not serious	Not serious	Strong association All plausible residual confounding would reduce demonstrated effect	⊕⊕○○ Low
CARDIOMETABOLIC ASSOCIATIONS						
Insulin resistance and insulin secretion dynamics						
HOMA-IR	Very serious	Serious	Not serious	Not serious	Strong association Plausible confounding would suggest spurious association	⊕○○○ Very low
Fasting insulin	Very serious	Serious	Not serious	Not serious	Strong association Plausible confounding would suggest spurious association	⊕○○○ Very low
AUC insulin	Very serious	Serious	Not serious	Serious	Strong association All plausible residual confounding would reduce demonstrated effect	⊕○○○ Very low
Insulin sensitivity index	Very serious	Serious	Not serious	Serious	All plausible residual confounding would reduce demonstrated effect	⊕○○○ Very low
Disorders of glucose tolerance						
Fasting glucose	Very serious	Serious	Not serious	Not serious	Strong association Plausible confounding would suggest spurious association	⊕○○○ Very low
AUC glucose	Very serious	Not serious	Not serious	Serious	All plausible residual confounding would reduce demonstrated effect	⊕○○○ Very low
Diabetes	Very serious	Serious	Not serious	Not serious	Strong association All plausible residual confounding would reduce demonstrated effect	⊕○○○ Very low

(continued)

Table 2. (continued)

Outcome	Risk of bias	Inconsistency of results	Indirectness of evidence	Imprecision	Other considerations	Quality of evidence
Atherogenic dyslipidaemia						
Total cholesterol	Not serious	Not serious	Not serious	Very serious	None	⊕○○○ Very low
HDL-cholesterol	Not serious	Not serious	Not serious	Not serious	Strong association	⊕⊕⊕○ Moderate
LDL-cholesterol	Not serious	Not serious	Not serious	Serious	None	⊕○○○ Very low
Triglycerides	Very serious	Serious	Not serious	Serious	All plausible residual confounding would reduce demonstrated effect	⊕○○○ Very low
Dyslipidaemia	Not serious	Not serious	Serious	Not serious	None	⊕○○○ Very low
Hypertension						
Systolic blood pressure	Very serious	Serious	Not serious	Serious	All plausible residual confounding would reduce demonstrated effect	⊕○○○ Very low
Diastolic blood pressure	Not serious	Serious	Not serious	Very serious	None	⊕○○○ Very low
Hypertension	Very serious	Serious	Serious	Not serious	Strong association All plausible residual confounding would reduce demonstrated effect	⊕○○○ Very low
Surrogate markers of cardiovascular disease						
Carotid intima-media thickness	Not serious	Very serious	Not serious	Very serious	None	⊕○○○ Very low
Cardiovascular morbidity and mortality						
Cardiovascular disease	Very serious	Not serious	Not serious	Not serious	None	⊕○○○ Very low
Coronary artery disease	Not serious	Serious	Not serious	Very serious	All plausible residual confounding would reduce demonstrated effect	⊕○○○ Very low
Myocardial infarction	Serious	Not serious	Not serious	Serious	Strong association All plausible residual confounding would reduce demonstrated effect	⊕⊕○○ Low
Stroke	Very serious	Not serious	Not serious	Serious	Strong association All plausible residual confounding would reduce demonstrated effect	⊕○○○ Very low
MENOPAUSAL MORBIDITY AND GYNAECOLOGICAL CANCER						
Breast cancer	Serious	Not serious	Not serious	Not serious	All plausible residual confounding would reduce demonstrated effect	⊕⊕○○ Low

GRADE, grading of recommendations, assessment, development, and evaluations; AUC, area under the curve; HDL, high-density lipoprotein; HOMA-IR, homeostasis model assessment of insulin resistance, LDL, low-density lipoprotein.
Quality of evidence: *High quality*: further research is very unlikely to change our confidence in the estimate of effect; *moderate quality*: further research in likely to have an important impact on our confidence in the estimate of effect and may change the estimate; *low quality*: further research is very likely to have an important impact on our confidence in the estimate of effect and is likely to change the estimate; *very low quality*: we are uncertain about the estimate.

Clinical manifestations

The information about symptoms and signs of androgen excess in menopausal women is scarce and mostly derives from non-functional causes of androgen excess such as virilizing tumours and hyperplasias (Alpanes *et al.*, 2012). In menopausal women with PCOS, hirsutism was the most frequently reported sign of androgen excess (Dahlgren *et al.*, 1992; Krentz *et al.*, 2007; Schmidt *et al.*, 2011a; Gabrielli *et al.*, 2015; Persson *et al.*, 2023). In a Swedish series, the prevalence of hirsutism was 64% in women with PCOS compared to 9% in control women by age 70 years (Schmidt *et al.*, 2011a) and 33% compared to 4% by age 81 years (Forslund *et al.*, 2021); in a different series of Swedish women with a median age of 50 years, these figures were 42% and 3% (Persson *et al.*, 2023) whereas in Brazilian women with a mean age of 59 years, these figures were 80% and 15%, respectively (Gabrielli *et al.*, 2015). Differences among patients and controls reached statistical significance (Schmidt *et al.*, 2011a; Gabrielli *et al.*, 2015; Forslund *et al.*, 2021; Persson *et al.*, 2023). Also, peri- and menopausal women with PCOS had higher hirsutism scores compared with control women (Alsamarai *et al.*, 2009). Acne, female pattern hair loss, or alopecia were also more common in another Swedish series of women with PCOS aged ≥45 years (Persson *et al.*, 2023).

Ultrasound examination of postmenopausal women showed that, as happens in premenopausal women, polycystic ovaries at this age were also associated with hirsutism, increased testosterone concentrations and increased ovarian size (Birdsall and Farquhar, 1996). Importantly, both ovarian volume and follicle number decreased with age both in women with PCOS and in controls, yet these indexes of polycystic ovarian morphology were larger in women with PCOS, regardless of age (Alsamarai *et al.*, 2009). In perimenopausal women with PCOS, oligomenorrhoea tended to ameliorate in the years preceding menopause (Dahlgren *et al.*, 1992; Pinola *et al.*, 2015). Menopause occurred at later ages in women with PCOS compared with controls in some studies (Li *et al.*, 2016; Meun *et al.*, 2018; Forslund *et al.*, 2019).

Also, the proportion of women that had reached menopause at a certain age was reduced in women with PCOS compared with control women; in two series these figures were 29% and 30% in patients with PCOS compared with 59% and 56% in controls, respectively (Dahlgren *et al.*, 1994; Talbott *et al.*, 2008). Similarly, only 27% women with PCOS showing FSH concentrations above 50 U/L had entered menopause, compared with 60% of control women (Dahlgren *et al.*, 1992), and these figures were 7% and 60%, respectively, in another study (Forslund *et al.*, 2019). In a study addressing diseases that are common in perimenopausal women, the mean age of menopause was 56 years in those with a diagnosis of PCOS (Li *et al.*, 2016).

However, a tendency towards delayed menopause in women with PCOS compared with controls did not reach statistical significance in some studies (Talbott *et al.*, 2008; Schmidt *et al.*, 2011a; Krentz *et al.*, 2012) and age at menopause was similar in both groups of women in others (Anderson *et al.*, 1997; Cibula *et al.*, 2000; Markopoulos *et al.*, 2011, 2013; Armeni *et al.*, 2013; Guoqing *et al.*, 2016; Merz *et al.*, 2016; Forslund *et al.*, 2022b), although a larger proportion of smokers in the PCOS group, by impairing ovarian function, might have influenced the result in one of these studies (Merz *et al.*, 2016). However, in one study women with PCOS, who were younger than controls, reached menopause at an earlier age (Gabrielli *et al.*, 2015). Also, in another study (Anderson *et al.*, 1997), the proportion of surgical menopause was increased in women with PCOS (74%) compared with controls (30%).

Meta-analysis of studies showing age at natural menopause showed no statistically significant differences between women with PCOS and controls (8 studies; 405 women with PCOS; 25 402 controls; MD −0.20 (−1.93, 1.52); SMD 0.00; (−0.32, 0.32); Cohen's $U+0\%$; very low certainty of evidence; Fig. 2). We had to exclude several studies that reported the years passed since menopause, instead of showing the actual age at which menopause occurred (Krentz *et al.*, 2007; Markopoulos *et al.*, 2011, 2013; Armeni *et al.*, 2013; Guoqing *et al.*, 2016).

Many studies showed an association of PCOS with weight excess and/or adiposity in menopausal patients. Most of these studies reported mean BMI in the overweight or obesity ranges (Dahlgren *et al.*, 1992; Anderson *et al.*, 1997; Cibula *et al.*, 2000; Margolin *et al.*, 2005; Krentz *et al.*, 2007; Alsamarai *et al.*, 2009; Puurunen *et al.*, 2011; Polotsky *et al.*, 2012; Armeni *et al.*, 2013; Chan *et al.*, 2013; Markopoulos *et al.*, 2013; Snyder *et al.*, 2014; Gabrielli *et al.*, 2015; Pinola *et al.*, 2015; Echiburu *et al.*, 2016; Guoqing *et al.*, 2016; Merz *et al.*, 2016; Ollila *et al.*, 2016; Schmidt *et al.*, 2017; Meun *et al.*, 2018, 2020; Forslund *et al.*, 2019, 2021; Persson *et al.*, 2023) in peri- or postmenopausal women with PCOS, together with increased waist circumference and waist-to-hip ratio. Moreover, waist circumference was increased compared to control women even in normal weight women with PCOS (Schmidt *et al.*, 2011a; Markopoulos *et al.*, 2013; Echiburu *et al.*, 2016). BMI, waist circumference, and the waist-to-hip ratio worsened in parallel to the number of PCOS diagnostic criteria met by the patients (Gabrielli *et al.*, 2015). Only in some of the studies were women with PCOS similar to controls in terms of BMI (Dahlgren *et al.*, 1992; Anderson *et al.*, 1997; Cibula *et al.*, 2000; Markopoulos *et al.*, 2013; Echiburu *et al.*, 2016; Merz *et al.*, 2016; Schmidt *et al.*, 2017; Forslund *et al.*, 2021).

Meta-analysis of BMI showed increased values in patients with PCOS compared with controls (23 studies; 2074 women with PCOS; 44 679 controls; MD 2.8 (1.9, 3.7); SMD 0.57 (0.38, 0.75); Cohen's $U+22\%$; low certainty of evidence; Fig. 2). The waist circumference (14 studies; 1040 women with PCOS; 6652 controls; MD 7.5 (5.1, 9.8); SMD 0.64 (0.42, 0.86); Cohen's $U+25\%$; moderate certainty of evidence; Fig. 2) and the waist-to-hip ratio (15 studies; 1249 women with PCOS; 37 806 controls; MD 0.03 (0.01, 0.04); SMD 0.38 (0.14, 0.61); Cohen's $U+15\%$; very low certainty of evidence; Fig. 2) were similarly increased in women with PCOS compared with controls.

Circulating androgen profiles

As occurs in premenopausal women (Escobar-Morreale *et al.*, 2008; Elhassan *et al.*, 2018), PCOS was the most common cause of androgen excess among postmenopausal women (Elhassan *et al.*, 2018). Severe hyperandrogenaemia was rare in postmenopausal women with PCOS, whose androgen levels were much lower compared with those of postmenopausal women presenting with androgen-secreting tumours or ovarian hyperthecosis (Elhassan *et al.*, 2018).

The most frequent findings in peri- and postmenopausal women with PCOS, compared with similarly aged women without PCOS, were statistically significant increases in circulating TT (Dahlgren *et al.*, 1992; Winters *et al.*, 2000; Talbott *et al.*, 2004b; Alsamarai *et al.*, 2009; Puurunen *et al.*, 2009; Markopoulos *et al.*, 2011, 2013; Polotsky *et al.*, 2012, 2014; Armeni *et al.*, 2013; Chan *et al.*, 2013; Gabrielli *et al.*, 2015; Ollila *et al.*, 2017; Meun *et al.*, 2018, 2020), FAI (Dahlgren *et al.*, 1992; Krentz *et al.*, 2007; Puurunen *et al.*, 2009, 2011; Markopoulos *et al.*, 2011; Schmidt *et al.*, 2011a; Armeni *et al.*, 2013; Polotsky *et al.*, 2014; Gabrielli *et al.*, 2015; Guoqing *et al.*, 2016; Ollila *et al.*, 2017; Meun *et al.*, 2018, 2020;

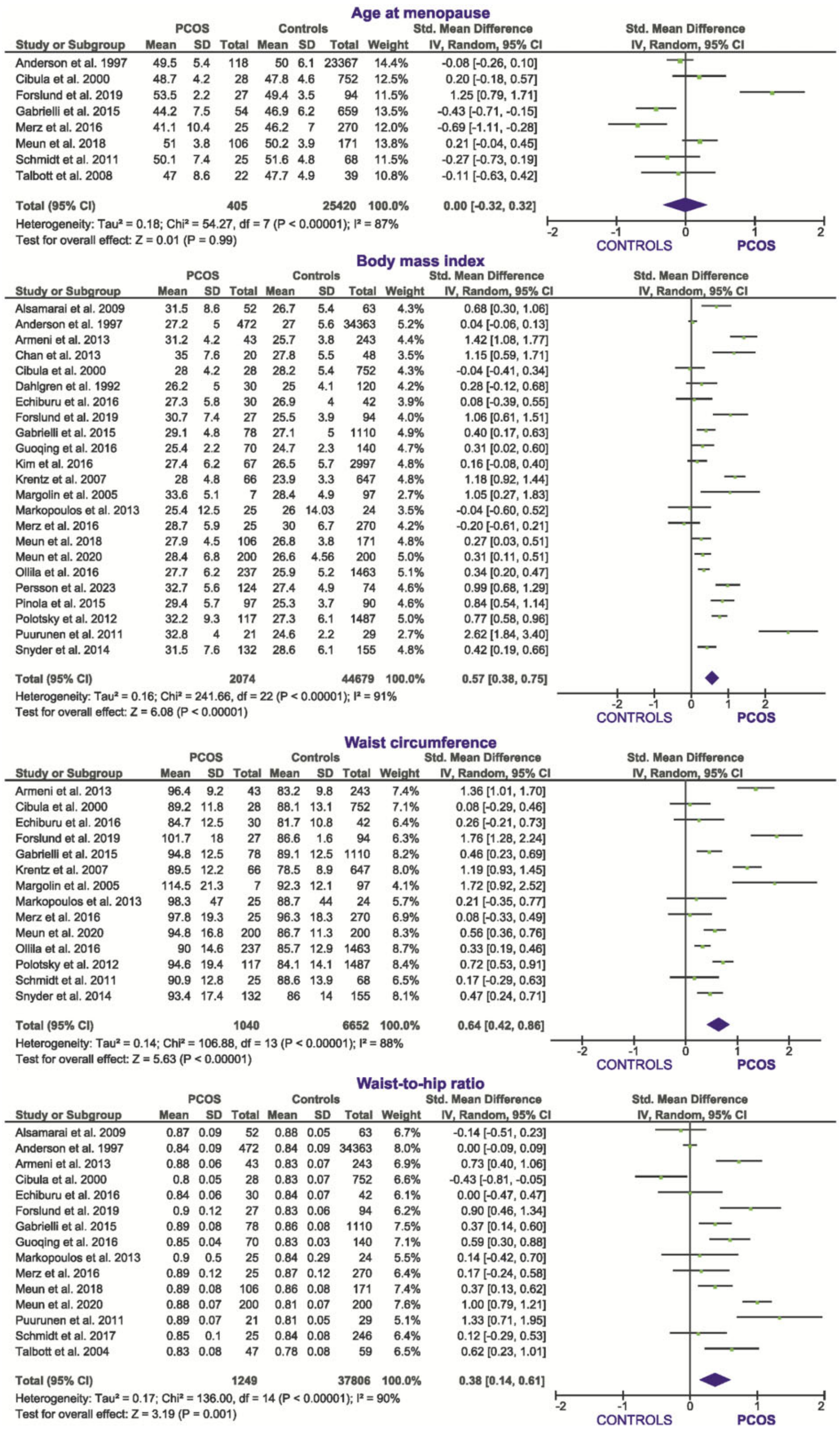


Figure 2. Meta-analysis (random effects model) of the standardized mean difference in age at menopause and anthropometric parameters between women with PCOS and controls. Funnel plots are depicted in Supplementary Fig. S1, except when less than 10 studies were included.

Persson *et al.*, 2023), and androstenedione (Dahlgren *et al.*, 1992; Alsamarai *et al.*, 2009; Puurunen *et al.*, 2009; Markopoulos *et al.*, 2011, 2013; Pinola *et al.*, 2015; Meun *et al.*, 2018) together with statistically significant decreases in serum concentrations of SHBG (Dahlgren *et al.*, 1992; Krentz *et al.*, 2007; Alsamarai *et al.*, 2009; Markopoulos *et al.*, 2011, 2013; Puurunen *et al.*, 2011; Schmidt *et al.*, 2011a; Armeni *et al.*, 2013; Gabrielli *et al.*, 2015; Pinola *et al.*, 2015; Meun *et al.*, 2018, 2020; Persson *et al.*, 2023).

Several studies also compared serum androgens in women with PCOS of different reproductive ages (Winters *et al.*, 2000; Puurunen *et al.*, 2011; Schmidt *et al.*, 2011a; Gabrielli *et al.*, 2015; Pinola *et al.*, 2015). Cross-sectional comparisons showed similar serum TT concentrations in pre- and postmenopausal patients with PCOS in one study (Gabrielli *et al.*, 2015), but serum androgens levels decreased—up to a 50% in some cases—in both patients with PCOS and controls after age 47 years in another study (Winters *et al.*, 2000; Schmidt *et al.*, 2011a), even though androgen levels remained increased in patients with PCOS compared with controls, regardless of menopause (Winters *et al.*, 2000; Schmidt *et al.*, 2011a). A prospective study with a 21-year follow-up confirmed the decrease in most serum androgens in postmenopausal women with and without PCOS (Schmidt *et al.*, 2011a). The decrease in TT, FAI, and DHEAS was larger in women with PCOS, to the extent that TT and DHEAS concentrations were no longer increased compared with those of controls (Schmidt *et al.*, 2011a). On the contrary, androstenedione increased in the controls and, together with FAI values, remained increased after menopause in women with PCOS compared with those of controls, in parallel with decreased SHBG concentrations (Schmidt *et al.*, 2011a). In fact, the TT to oestradiol and androstenedione to oestradiol ratios were increased in menopausal women with PCOS compared with controls in one study (Puurunen *et al.*, 2011) and increased TT (Markopoulos *et al.*, 2011), calculated free testosterone, FAI, and androstenedione were the best predictors of PCOS at all ages—including menopause—in another report (Pinola *et al.*, 2015).

Other studies did not find differences between menopausal women with or without PCOS in the circulating concentrations of TT (Talbott *et al.*, 2004a, 2008; Krentz *et al.*, 2007; Puurunen *et al.*, 2011; Schmidt *et al.*, 2011a; Pinola *et al.*, 2015; Yin *et al.*, 2018; Forslund *et al.*, 2020; Persson *et al.*, 2023), androstenedione (Schmidt *et al.*, 2011a; Armeni *et al.*, 2013), and SHBG (Guoqing *et al.*, 2016; Forslund *et al.*, 2020), as well as in FAI values (Pinola *et al.*, 2015; Forslund *et al.*, 2020). In fact, no differences were found in hormonal variables of menopausal women with and without PCOS in another report (Margolin *et al.*, 2005); however, this study compared only seven patients with PCOS with 97 control women, and almost half of patients with PCOS were receiving metformin, an insulin sensitizer drug that may decrease serum androgen levels by ameliorating insulin resistance (Naderpoor *et al.*, 2016). Moreover, in a large study, androstenedione levels were actually decreased in patients with PCOS compared with controls (Meun *et al.*, 2020).

Three reports (Puurunen *et al.*, 2009, 2011; Markopoulos *et al.*, 2011) addressed the dynamics of the hypothalamic–pituitary-adrenal and ovarian axes and the adrenal and ovarian steroid secretion profiles of menopausal women with PCOS (Table 3), and another article reported increased 17-hydroxyprogesterone concentrations in these patients (Markopoulos *et al.*, 2013). These authors concluded that, although androgen secretion decreases with age in control women, androgen excess may persist after menopause in women with PCOS, with the adrenals contributing substantially to this increase (Puurunen *et al.*, 2009, 2011; Markopoulos *et al.*, 2011, 2013).

Table 3. Dynamic examination of the hypothalamic–pituitary–adrenal and ovarian axes and adrenal and ovarian steroid secretion profiles in peri- and postmenopausal women with PCOS.

Markopoulos *et al.*, 2011	Puurunen *et al.*, 2009	Puurunen *et al.*, 2011
Basal concentrations		
↑ TT, FAI, Δ^4A, DHEAS, P4, and 17-OHP in PCOS, independent of BMI	↓ 17-OHP with age regardless of PCOS influenced by BMI	↓ Δ^4A with age, but remains ↑ in PCOS
No differences in urine cortisol	Cortisol, Δ^4A, and DHEAS: decreases with age in controls but remain stable in PCOS	
Corticotrophin-releasing hormone test		
No differences in pituitary function		
Cosyntropin (1-24corticotrophin) test		
↑ TT, Δ^4A, and DHEAS levels in PCOS that result from increased basal levels (no net increment in secretion)	↑ AUC of TT, Δ^4A, DHEAS, and 17-OHP in PCOS	
Normal cortisol and 17-OHP in PCOS	↑ AUC of TT in menopausal PCOS compared with premenopausal PCOS. ↑ AUC of cortisol in PCOS compared with controls regardless of menopause, with larger ↑ in menopausal patients	
Dexamethasone suppression test		
Cortisol, TT, FAI, Δ^4A, DHEAS, and 17-OHP decrease in PCOS and controls		
↑ FAI, Δ^4A, and 17-OHP in PCOS		
No changes in LH in any group		
HCG test		
		↑ 17-OHP and Δ^4A in PCOS ↑ 17-OHP and Δ^4A in premenopausal PCOS compared with menopausal PCOS AUC of T remains unchanged after menopause
Interpretation		
Androgen excess is mostly adrenal in menopausal women with PCOS	Androgen production decreases with age	Androgen synthesis decreases with age but remains increased in PCOS compared with controls
Androgen synthesis is increased compared with controls	Adrenal steroid synthesis remains increased in women with PCOS during the menopausal transition	Ovarian secretion of T remains stable during the menopausal transition
The ovary of menopausal women with PCOS might secrete 17-OHP and Δ^4A in excess		The contribution of the menopausal ovary to androgen excess is minimal

↑, increase; ↓, decrease; Δ^4A, androstenedione; 17-OHP, 17-hydroxyprogesterone; DHEAS, dehydroepiandrosterone-sulphate; FAI, free androgen index; HCG, human chorionic gonadotrophin; P4, progesterone; T, testosterone; TT, total testosterone.

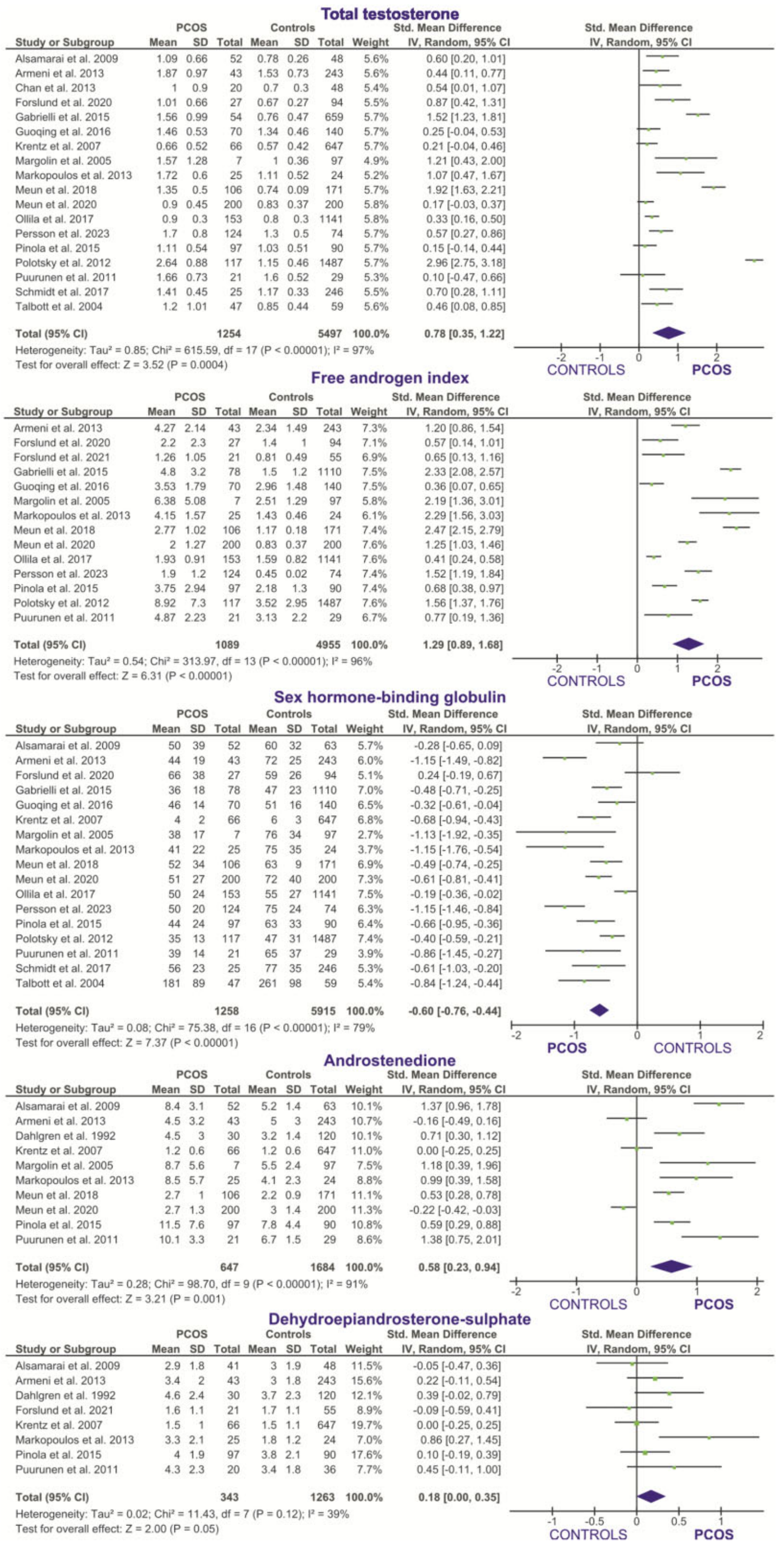


Figure 3. Meta-analysis (random effects model) of the standardized mean difference in serum androgen concentrations between women with PCOS and controls. Funnel plots are depicted in Supplementary Fig. S1, except when less than 10 studies were included.

Meta-analysis of available studies showed that peri- and postmenopausal women with PCOS, compared with controls, had increased serum TT concentrations (18 studies; 1254 women with PCOS; 5497 controls; MD 0.4 (0.2, 0.5); SMD 0.78 (0.35, 1.22); Cohen's $U+30\%$; very low certainty of evidence; Fig. 3), FAI (14 studies; 1089 women with PCOS; 4995 controls; MD 1.66 (1.22, 2.10); SMD 1.29 (0.89, 1.68); Cohen's $U+48\%$; low certainty of evidence; Fig. 3), and androstenedione (10 studies; 647 women with PCOS; 1684 controls; MD 1.24 (0.65, 1.84); SMD 0.58 (0.23, 0.94); Cohen's $U+23\%$; very low certainty of evidence; Fig. 3), whereas SHBG was reduced in the former (17 studies; 1258 women with PCOS; 5915 controls; MD −17 (−22, −11); SMD −0.60 (−0.76, −0.44); Cohen's U −24%; low certainty of evidence; Fig. 3). There was also a near-significant trend towards an increase in

DHEAS in women with PCOS (8 studies; 343 women with PCOS; 1263 controls; MD 0.31 (−0.02, 0.64); SMD 0.18 (0.00, 0.35); Cohen's $U+7\%$, very low certainty of evidence; Fig. 3).

When appropriate, we excluded from the meta-analyses studies that described patients also included in a larger report of the same cohorts (Dahlgren *et al.*, 1992, 1994; Talbott *et al.*, 2000, 2004b, 2008; Puurunen *et al.*, 2009; Markopoulos *et al.*, 2011; Schmidt *et al.*, 2011a,b; 2017; Forslund *et al.*, 2021; Kujanpää *et al.*, 2022; Polonen *et al.*, 2022; Forslund *et al.*, 2022b, 2023), and studies that did not report results as mean and SD and we could not estimate these statistics from the original data, and/or we did not obtain any response to our queries for data clarification (Yin *et al.*, 2018).

Cardiometabolic associations

Insulin resistance and insulin secretion dynamics

Most studies used HOMA-IR as a surrogate marker of insulin resistance in the fasting state (Krentz *et al.*, 2007; Alsamarai *et al.*, 2009; Puurunen *et al.*, 2011; Schmidt *et al.*, 2011a; Polotsky *et al.*, 2012; Armeni *et al.*, 2013; Chan *et al.*, 2013; Markopoulos *et al.*, 2013; Gabrielli *et al.*, 2015; Echiburu *et al.*, 2016; Guoqing *et al.*, 2016; Merz *et al.*, 2016; Forslund *et al.*, 2020, 2022b; Meun *et al.*, 2020). HOMA-IR was increased in menopausal women with PCOS compared with controls in some studies (Krentz *et al.*, 2007; Alsamarai *et al.*, 2009; Puurunen *et al.*, 2011; Polotsky *et al.*, 2012; Armeni *et al.*, 2013; Chan *et al.*, 2013; Gabrielli *et al.*, 2015; Forslund *et al.*, 2020). In one of these studies, HOMA-IR increased in women with PCOS during the menopausal transition in parallel with the number of diagnostic criteria met by the patients (Gabrielli *et al.*, 2015). In another four reports, HOMA-IR was increased in both in menopausal controls and in women with PCOS (Schmidt *et al.*, 2011a; Echiburu *et al.*, 2016; Meun *et al.*, 2020; Forslund *et al.*, 2022b) whereas this index of insulin resistance was normal in menopausal women with PCOS in three studies (Markopoulos *et al.*, 2013; Guoqing *et al.*, 2016; Merz *et al.*, 2016).

Meta-analysis of available studies showed increased HOMA-IR in menopausal women with PCOS compared with controls (15 studies; 952 women with PCOS; 5606 controls; MD 0.90 (0.50, 1.30); SMD 0.56 (0.27, 0.84); Cohen's $U+22\%$; very low certainty of evidence; Fig. 4). In agreement with the increase in HOMA-IR, meta-analysis of fasting insulin showed increased concentrations in women with PCOS compared with controls (14 studies; 994 women with PCOS; 5403 controls; MD 23 (14, 32); SMD 0.61 (0.38, 0.83); Cohen's $U+24\%$; very low certainty of evidence; Fig. 4).

Regarding the postprandial phase, several studies assessed insulin resistance and dynamics during an oral glucose tolerance test. The AUC of insulin during the oral glucose tolerance test was increased in menopausal women with PCOS compared with controls (Puurunen *et al.*, 2011; Markopoulos *et al.*, 2013; Echiburu *et al.*, 2016; Ollila *et al.*, 2017), even in those showing normal HOMA-IR values (Markopoulos *et al.*, 2013). In one of these studies, these differences were only present in overweight and obese individuals (Ollila *et al.*, 2017). In another report (Puurunen *et al.*, 2011), the AUC of insulin was increased, and the composite insulin sensitivity index was decreased, in pre- and postmenopausal patients with PCOS. Moreover, menopause resulted into a worsening of these indexes of insulin resistance in patients with PCOS, and such changes occurred regardless of BMI (Puurunen *et al.*, 2011). On the contrary, a later report found an increased AUC of insulin and first- and second-phase insulin secretion indexes in women with PCOS compared with controls without changes in the insulin sensitivity index, and in the AUC of glucose during the oral glucose tolerance test (Markopoulos *et al.*, 2013). A positive correlation of FAI with fasting insulin, AUC of insulin and glucose and HOMA-IR was found (Markopoulos *et al.*, 2013). These authors concluded that early postmenopausal women with PCOS are characterized by hyperinsulinaemia but attenuated insulin resistance, with amelioration of androgen excess possibly contributing to the latter (Markopoulos *et al.*, 2013).

Meta-analysis of AUC of insulin as an index of insulin dynamics during the oral glucose tolerance test showed increased concentrations in women with PCOS compared with controls (4 studies; 229 women with PCOS; 1236 controls; MD 15 782 (3282, 28 282); SMD 0.55 (0.13, 0.97); Cohen's $U+22\%$; very low certainty of evidence; Fig. 4). Yet meta-analysis of the composite insulin sensitivity index derived from glucose and insulin concentrations during the oral glucose tolerance test (Matsuda and DeFronzo, 1999) showed no differences between women with PCOS and controls (3 studies; 204 women with PCOS; 1212 controls; MD −1.6 (−3.5, 0.4); SMD −0.33 (−0.74, 0.08); Cohen's U −13%; very low certainty of evidence; Fig. 4), reflecting the discrepancies in the results of different reports consisting of decreased (Ollila *et al.*, 2017) or similar (Markopoulos *et al.*, 2015; Echiburu *et al.*, 2016) values in women with PCOS compared with controls. Another report, that calculated the insulin sensitivity index differently, showed also no differences among patients with PCOS and controls (Markopoulos *et al.*, 2013).

Adipokines

The Rancho Bernardo Study reported data regarding the adipokine profiles of pre- and postmenopausal women with PCOS (Krentz and Barrett-Connor, 2009; Krentz *et al.*, 2012). In postmenopausal women, the PCOS phenotype consisted of the presence of three or more characteristics, such as a personal history of oligomenorrhoea, history of infertility or miscarriage, current or past clinical or hormonal evidence of hyperandrogenism, central obesity, and/or biochemical evidence of insulin resistance (Krentz and Barrett-Connor, 2009; Krentz *et al.*, 2012). Similar to their premenopausal counterparts, menopausal women with PCOS showed increased circulating leptin concentrations and decreased adiponectin and ghrelin levels compared with controls, changes that were proportional to the number of PCOS criteria matched by the patients (Krentz and Barrett-Connor, 2009; Krentz *et al.*, 2012). Only the changes in adiponectin were independent of BMI and waist circumference, with leptin levels reflecting the overall adiposity of these women (Krentz *et al.*, 2012).

In one of the studies mentioned above that addressed fasting and post-load insulin resistance, menopausal women with PCOS showed no changes in comparisons to controls in fasting adiponectin, leptin, visfatin, and retinol binding protein-4 (Markopoulos *et al.*, 2013). Only the lipocalin-2 to waist circumference ratio was increased in menopausal women with PCOS, while the FAI correlated with visfatin (Markopoulos *et al.*, 2013).

Disorders of glucose tolerance

In several studies, fasting glucose was higher and/or impaired fasting glucose (5.6–6.9 mmol/L) was more frequent, in menopausal women with PCOS compared with controls, putting the former at risk for the development of diabetes (Krentz *et al.*, 2007; Alsamarai *et al.*, 2009; Polotsky *et al.*, 2012; Chan *et al.*, 2013; Gabrielli *et al.*, 2015; Meun *et al.*, 2018, 2020; Forslund *et al.*, 2020). Impaired fasting glucose was frequent in menopausal women regardless of PCOS in two studies (Cibula *et al.*, 2000; Merz *et al.*, 2016), and in others fasting glycaemia was unaltered in these women (Talbott *et al.*, 2008; Puurunen *et al.*, 2011; Armeni *et al.*,

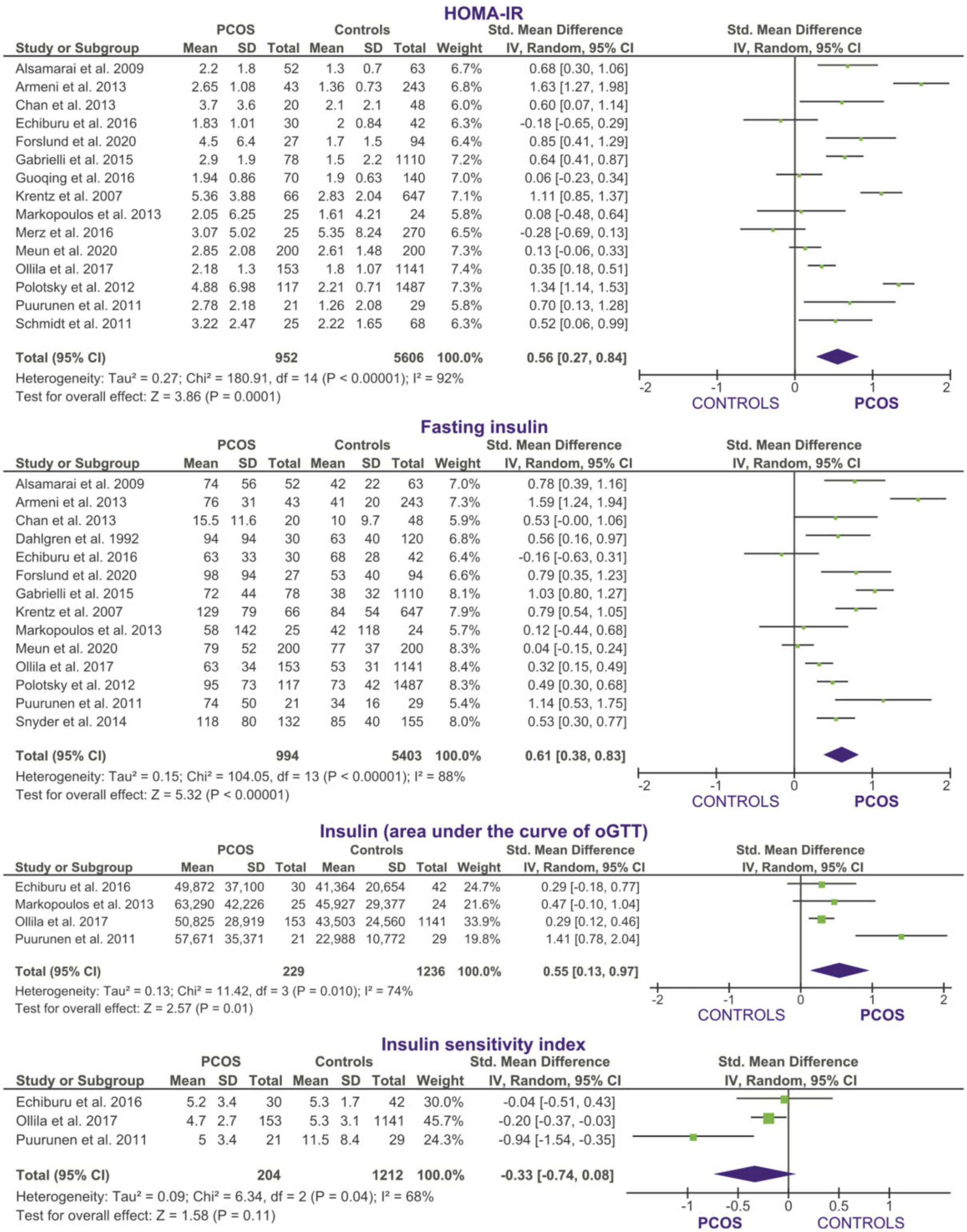


Figure 4. Meta-analysis (random effects model) of the standardized mean difference in surrogate indexes of insulin resistance and dynamics between women with PCOS and controls. Funnel plots are depicted in Supplementary Fig. S1, except when less than 10 studies were included. oGTT, oral glucose tolerance test.

2013; Markopoulos *et al.*, 2013; Echiburu *et al.*, 2016; Guoqing *et al.*, 2016; Ollila *et al.*, 2017), even in the presence of increased fasting insulin concentrations (Puurunen *et al.*, 2011; Armeni *et al.*, 2013; Markopoulos *et al.*, 2013). Compared with controls, the AUC of glucose during the oral glucose tolerance test was increased in women with PCOS in some reports (Puurunen *et al.*, 2011; Ollila *et al.*, 2017), but not in others (Markopoulos *et al.*, 2013; Echiburu *et al.*, 2016).

Meta-analysis of fasting glucose concentrations showed increased values in patients with PCOS compared with controls (17 studies; 1148 women with PCOS; 6544 controls; MD 0.4 (0.3, 0.5); SMD 0.48 (0.29, 0.68); Cohen's $U+19\%$; very low certainty of evidence; Fig. 5), whereas the AUC of glucose during the oral glucose tolerance test was not different among these groups (4 studies; 229 women with PCOS; 1236 controls; MD 43 (−5, 91); SMD 0.25 (−0.01, 0.50); Cohen's $U+10\%$; very low certainty of evidence;

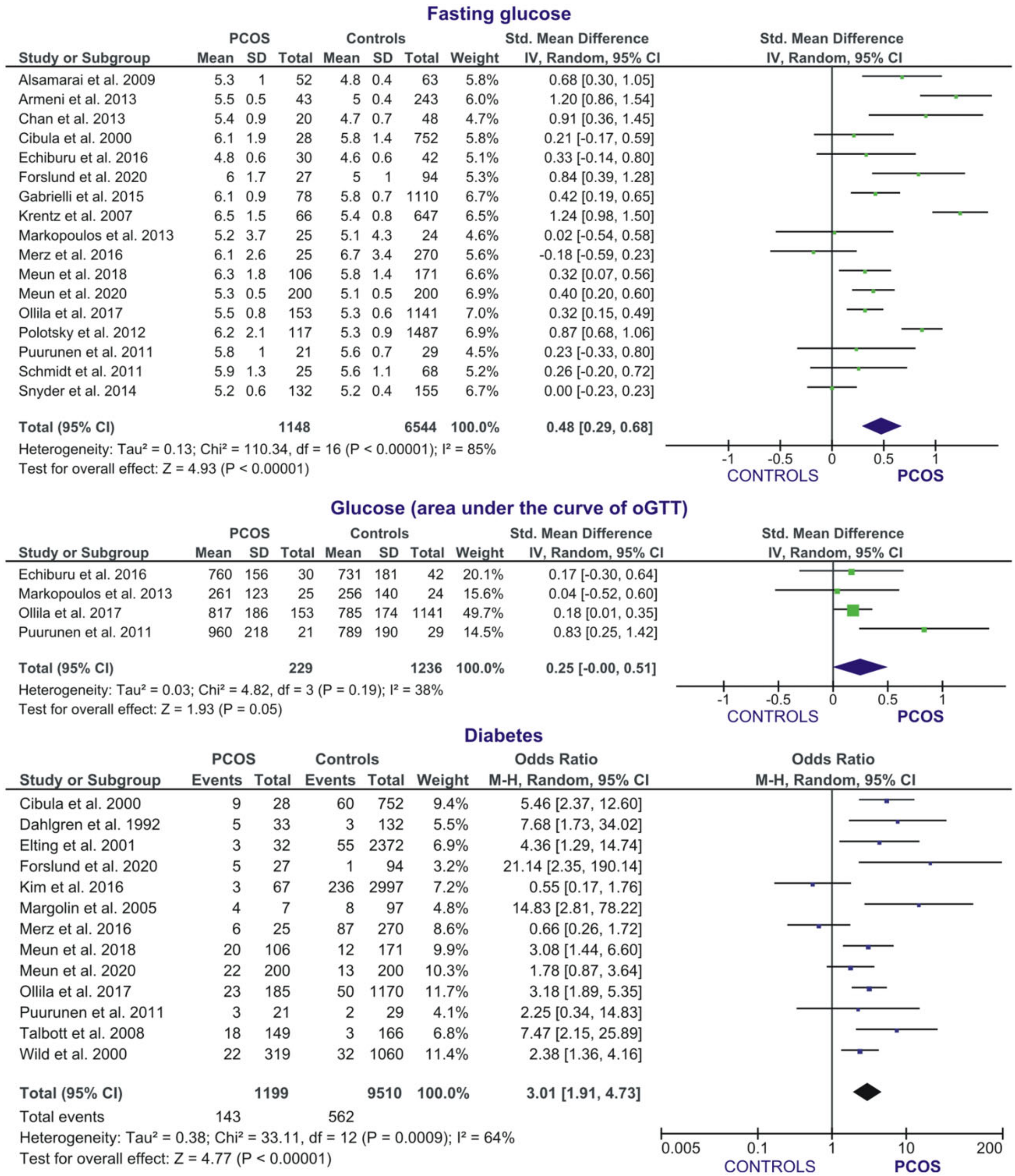


Figure 5. Meta-analysis (random effects model) of the standardized mean difference in fasting and post-load glucose levels and in the odds-ratio for diabetes, between women with PCOS and controls. Funnel plots are depicted in Supplementary Fig. S1, except when less than 10 studies were included. oGTT, oral glucose tolerance test.

Fig. 5). Moreover, meta-analysis of reports describing the occurrence of type 2 diabetes—or drug treatment for diabetes—in women with PCOS compared with controls showed an increased OR in the former (13 studies; 1199 women with PCOS; 9510 controls; OR 3.01 (1.91, 4.73); very low certainty of evidence; Fig. 5).

In one study, waist circumference >88 cm and more severe grades of hyperandrogenaemia and insulin resistance were more frequent in patients with diabetes compared to those without diabetes among women with PCOS (Krentz *et al.*, 2007). In another study, an inherited basis for cardiometabolic disease was suggested by the finding of cardiovascular disease and/or diabetes in at least one of the parents of women with PCOS (Dahlgren *et al.*, 1992).

Atherogenic dyslipidaemia

Atherogenic dyslipidaemia, consisting of decreased concentrations of HDL-cholesterol and increased TG, was a common finding in reports of peri- and postmenopausal women with PCOS

(Margolin et al., 2005; Krentz et al., 2007; Alsamarai et al., 2009; Polotsky et al., 2012; Armeni et al., 2013; Merz et al., 2016; Meun et al., 2018), with higher TG concentrations associating with polycystic ovaries in another report (Birdsall and Farquhar, 1996). Other authors did not find atherogenic dyslipidaemia but increased LDL-cholesterol concentrations in these women (Guoqing et al., 2016).

Several reports did not find differences in the lipid profile among peri- and postmenopausal women with or without PCOS or polycystic ovaries (Talbott et al., 1998; Cibula et al., 2000; Schmidt et al., 2011b; Chan et al., 2013; Gabrielli et al., 2015; Echiburu et al., 2016; Meun et al., 2020). One of these studies showed that total cholesterol, TG, lipid accumulation product, and visceral adiposity index were increased in premenopausal women with PCOS compared with controls, but these differences were not present in menopausal women (Echiburu et al., 2016). In another, premenopausal women with PCOS presented with increased cholesterol concentrations, and postmenopausal patients with PCOS with increased TG levels, compared to their control counterparts (Gabrielli et al., 2015). There was a trend towards the lowest HDL-cholesterol levels being observed in those menopausal patients matching more criteria for PCOS that did not reach statistical significance when compared with the controls (Gabrielli et al., 2015).

Meta-analyses of the lipid profile confirmed the association of PCOS during the menopausal transition with atherogenic dyslipidaemia, consisting of decreased HDL-cholesterol concentrations (13 studies; 875 women with PCOS; 3909 controls; MD −0.13 (−0.18, −0.08); SMD −0.32 (−0.46, −0.19); Cohen's U −13%; moderate certainty of evidence; Fig. 6) and increased TG (13 studies; 875 women with PCOS; 3909 controls; MD 0.20 (0.09, 0.30); SMD 0.31 (0.16, 0.46); Cohen's $U+12\%$; very low certainty of evidence; Fig. 6).

On the contrary, total cholesterol (13 studies; 875 women with PCOS; 3909 controls; MD −0.01 (−0.10, 0.13); SMD 0.01 (−0.11, 0.14); Cohen's U −0.0%; very low certainty of evidence; Fig. 6) and LDL-cholesterol concentrations (11 studies; 749 women with PCOS; 3690 controls; MD 0.05 (−0.07, 0.19); SMD 0.07 (−0.07, 0.21); Cohen's $U+3\%$; very low certainty of evidence; Fig. 6), as well as the OR for dyslipidaemia or drug treatment for dyslipidaemia (7 studies; 994 women with PCOS; 4816 controls; OR 1.12 (0.72, 1.74); very low certainty of evidence; Fig. 6) were similar among women with PCOS and controls.

Hypertension

Several reports addressed the prevalence of hypertension in peri- and postmenopausal women with or without PCOS (Dahlgren et al., 1992; Cibula et al., 2000; Wild et al., 2000; Elting et al., 2001; Margolin et al., 2005; Talbott et al., 2008; Schmidt et al., 2011b; Polotsky et al., 2012; Armeni et al., 2013; Echiburu et al., 2016; Kim et al., 2016; Merz et al., 2016; Ollila et al., 2016, 2019; Meun et al., 2020; Forslund et al., 2022a,b). Another study was excluded from meta-analysis of frequency of hypertension and blood pressure values because women with PCOS and controls were matched for prevalence of hypertension (Guoqing et al., 2016). Reports were discordant for an increase in the prevalence of hypertension in women with PCOS compared with controls. Moreover, other studies described increased mean blood pressure values in menopausal patients regardless of PCOS status (Krentz et al., 2007; Polotsky et al., 2012, 2014; Merz et al., 2016).

Meta-analysis of the prevalence of hypertension revealed an increased OR for having hypertension—or being on drug treatment for hypertension—in women with PCOS compared with controls (14 studies; 1284 women with PCOS; 11 240 controls; OR 1.79 (1.36, 2.36); very low certainty of evidence; Fig. 7). Also, perimenopausal women with self-reported PCOS were more likely to be on antihypertensive drugs compared with controls by age 46 years (Ollila et al., 2019) and the same occurred in postmenopausal women with the syndrome (Armeni et al., 2013).

However, our meta-analyses showed that SBP was only mildly increased in peri- and postmenopausal women with PCOS compared with controls (13 studies; 1042 women with PCOS; 5232 controls; MD 2 (−0, 4); SMD 0.14 (0.03, 0.26); very low certainty of evidence; Cohen's $U+6\%$; Fig. 7), and DBP was similar in the two groups (13 studies; 1042 women with PCOS; 5232 controls; MD 0 (−1, 2); SMD 0.04 (−0.09, 0.17); Cohen's $U+2\%$; very low certainty of evidence; Fig. 7).

Weight excess may play a major role in the association of PCOS with hypertension during the menopausal transition, as has been reported during the premenopausal years (Luque-Ramirez et al., 2007). By age 46 years, the OR for hypertension in women with PCOS was increased compared with controls regardless of BMI, yet the OR doubled in those patients presenting with weight excess (Ollila et al., 2019). In older women, however, the association of PCOS with hypertension depended mostly on the association of PCOS with weight excess (Armeni et al., 2013). Anyhow, hypertension associated unfavourable electrocardiographic changes, suggestive of septal and left-ventricular hyperplasia, in perimenopausal women regardless of PCOS (Ollila et al., 2019).

Chronic low-grade inflammation and prothrombotic status

Only a few reports included serum C reactive protein concentrations as a surrogate marker of low-grade chronic inflammation showing increased levels in women with PCOS compared with controls regardless of peri- or postmenopausal status (Puurunen et al., 2011; Markopoulos et al., 2013; Gabrielli et al., 2015). However, patients and controls were not matched for BMI in one of these studies. The increase in C reactive protein concentrations was proportional to the number of PCOS features of the patients (Gabrielli et al., 2015).

Regarding coagulation factors, circulating fibrinogen and plasminogen activator inhibitor-1 concentrations were similar, and increased similarly with age, among peri- and menopausal women with or without PCOS in one study (Schmidt et al., 2011b), yet fibrinogen and factor VII levels were reduced in patients with PCOS compared with controls in another report (Dahlgren et al., 1994).

Hypothyroidism

A single report suggested that hyperandrogenism might protect against the development of hypothyroidism in menopausal women with PCOS, even though the prevalence of thyroperoxidase autoantibodies was similar in this population and in appropriate controls (Schmidt et al., 2017). No difference in the ORs for hypothyroidism in peri- and postmenopausal women with PCOS and controls was found in another study (Forslund et al., 2022a), yet another study reported increased thyrotrophin concentrations in women with hyperandrogenaemia and oligo-ovulation compared with women with either androgen excess or oligo-ovulation alone, or compared with eumenorrhoeic women (Polotsky et al., 2012).

Surrogate markers of cardiovascular disease

Arterial stiffness, as measured by pulse wave velocity, was increased in postmenopausal women with PCOS compared with

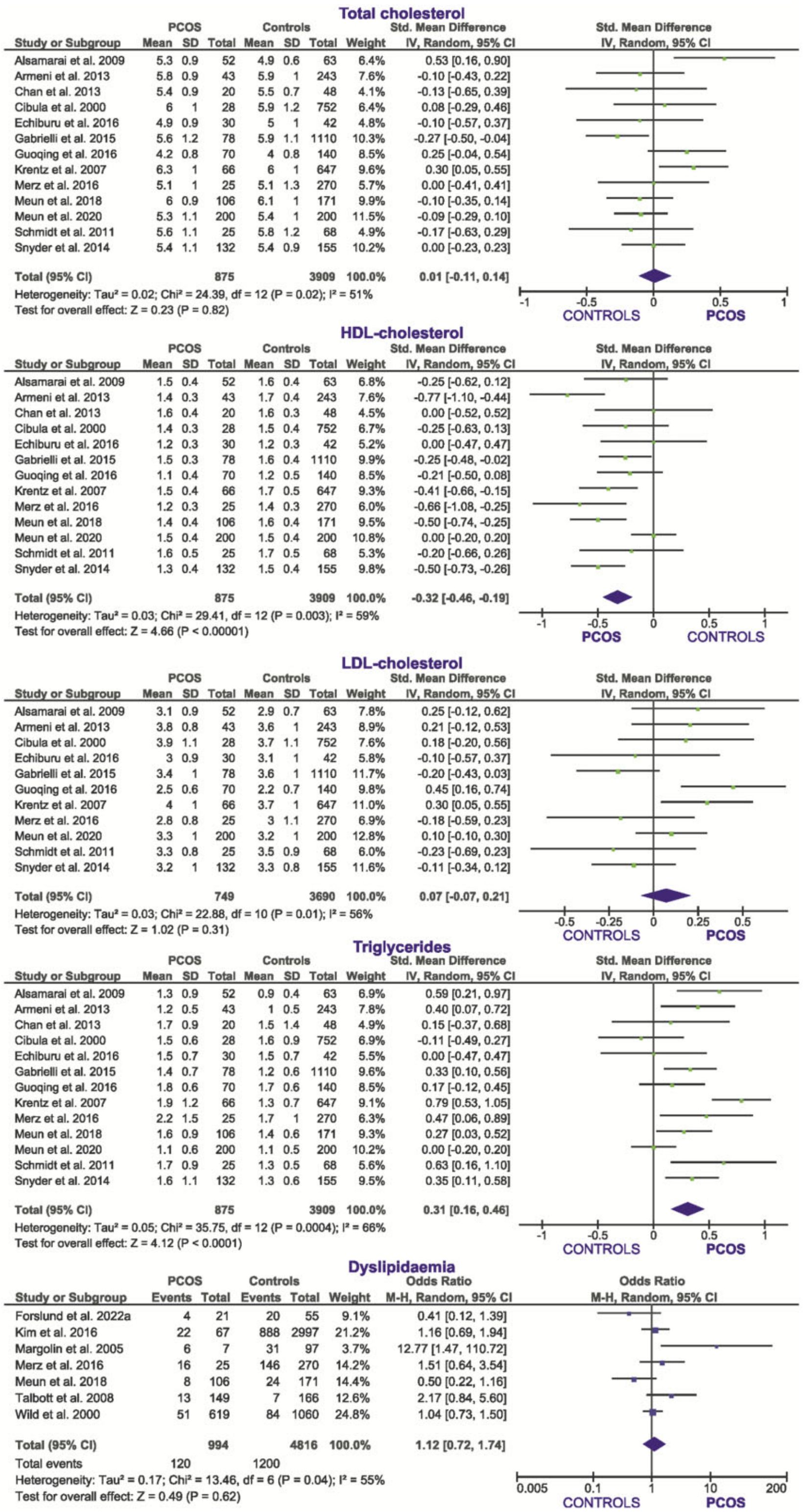


Figure 6. Meta-analysis (random effects model) of the standardized mean difference in serum lipids, and in the odds-ratio for dyslipidaemia, between women with PCOS and controls. Funnel plots are depicted in Supplementary Fig. S1, except when less than 10 studies were included.

controls, regardless of age, BMI, or blood pressure, with insulin resistance and androgen excess being major determinants of this finding (Armeni *et al.*, 2013). No differences among groups were found in other indexes of arterial function such as flow mediated dilation and augmentation index (Armeni *et al.*, 2013).

cIMT was measured in several studies as a marker of subclinical atherosclerosis, with discrepant results. cIMT was found to be increased (Talbott *et al.*, 2000), similar (Armeni *et al.*, 2013), or even decreased (Meun *et al.*, 2020) in perimenopausal and postmenopausal women with PCOS compared to controls. Meta-analysis of these data showed no statistically significant differences among patients and controls (3 studies; 290 women with PCOS; 503 controls; MD 0.0 (−0.1, 0.1); SMD −0.08 (−1.14, 0.99); Cohen's U −3%; very low certainty of evidence; Fig. 8). In another

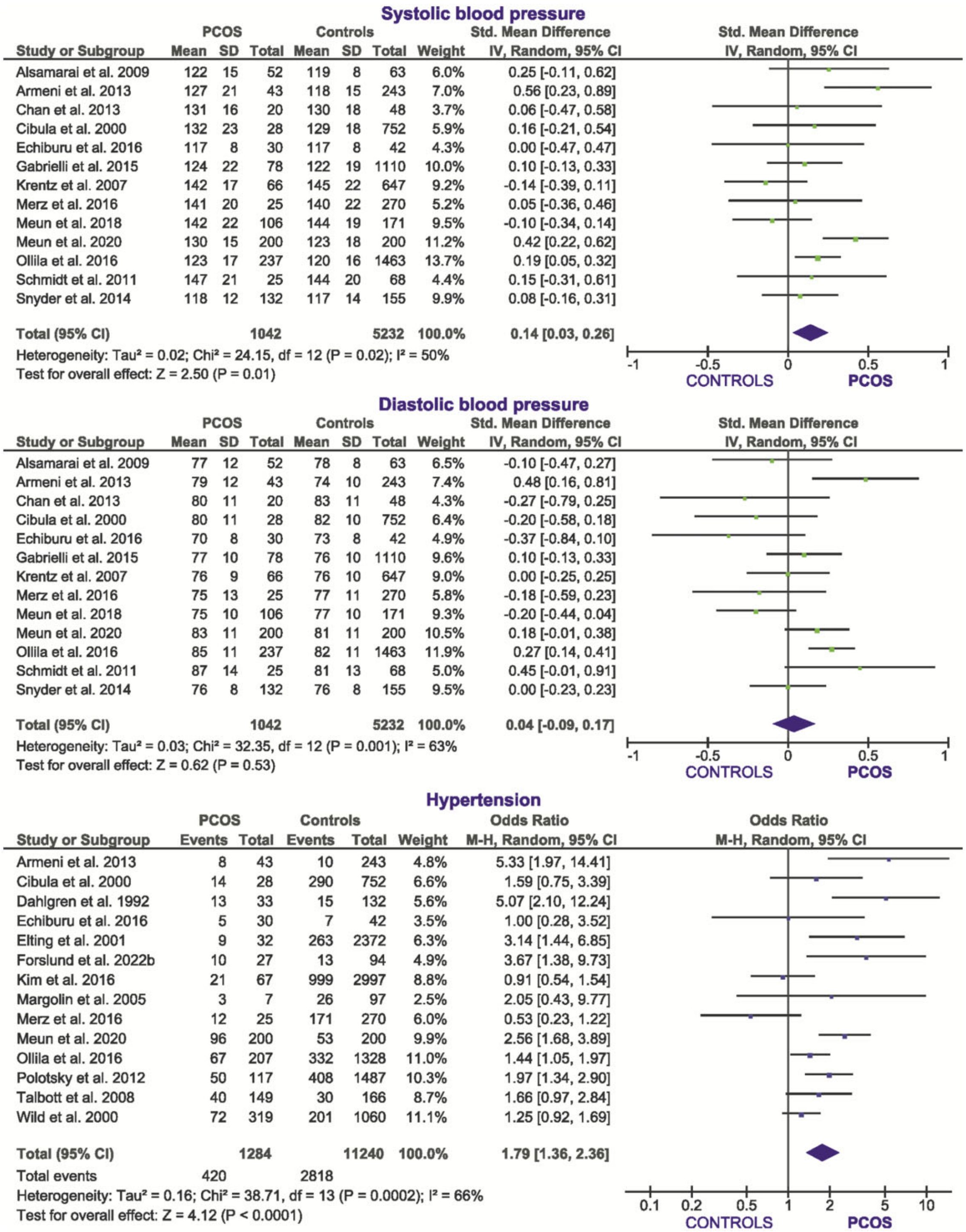


Figure 7. Meta-analysis (random effects model) of the standardized mean difference in blood pressure values, and in the odds-ratio for hypertension, between women with PCOS and controls. Funnel plots are depicted in Supplementary Fig. S1, except when less than 10 studies were included.

report, women in the highest quintiles of circulating dehydroepiandrosterone and androstenedione presented with the lowest cIMT values (Meun *et al.*, 2018).

Coronary artery calcification was more prevalent and more severe in peri- and postmenopausal women with PCOS than in controls in one report (Talbott *et al.*, 2008). Major determinants of coronary artery calcification were BMI in both women with PCOS and in controls, together with surgical menopause in the former and natural menopause in the latter (Talbott *et al.*, 2008). Moreover, in another study coronary angiography showed that, albeit coronary artery disease was similarly frequent in postmenopausal women with or without PCOS, multivessel disease tended to be more prevalent in women with the syndrome; however, such a difference did not reach statistical significance (Merz *et al.*, 2016). Finally, a single study addressed leukocyte telomere length as a possible risk factor for cardiometabolic disease in women with PCOS by age 46 years but found no differences with those of control women (Polonen *et al.*, 2022).

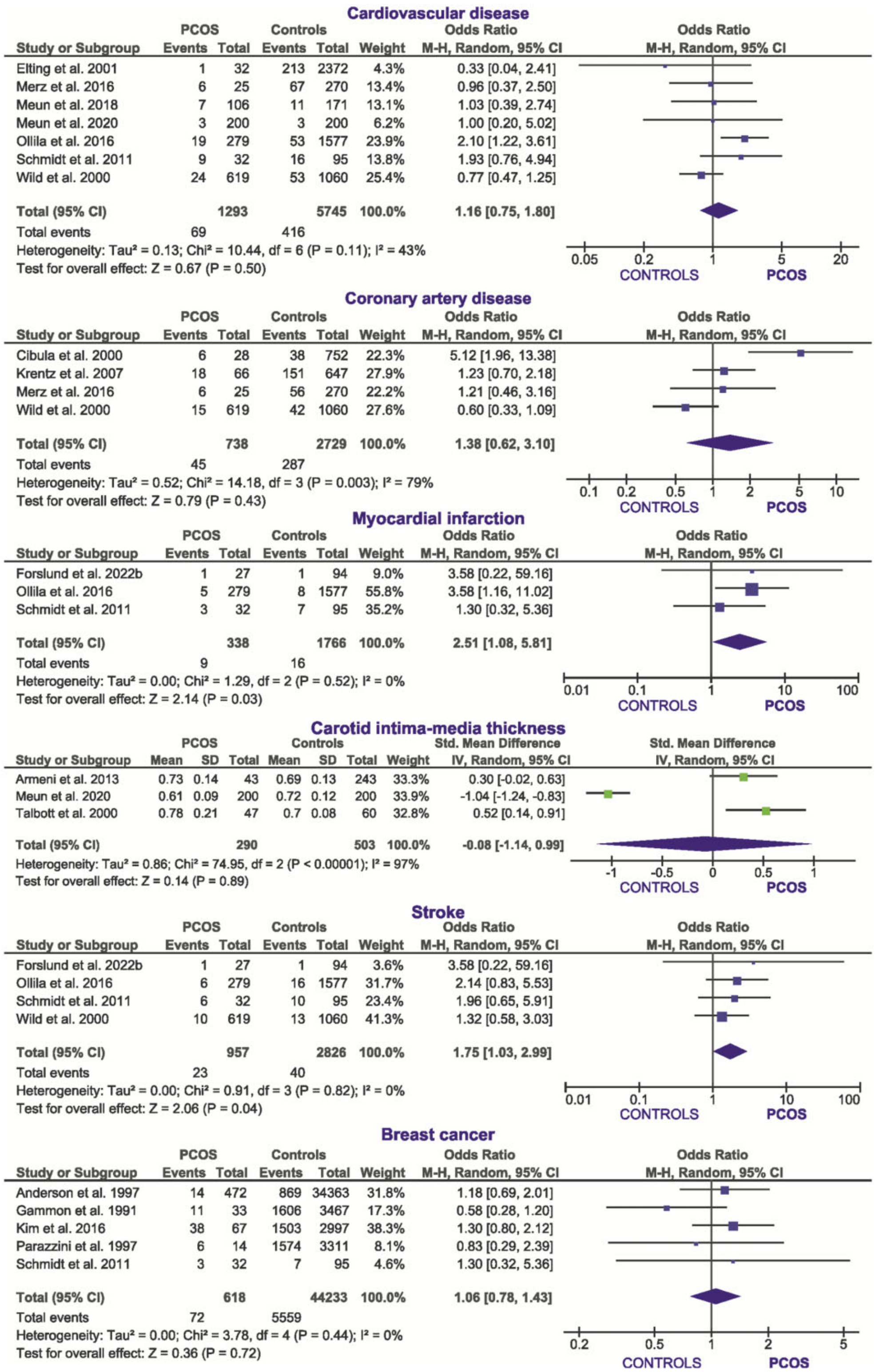


Figure 8. Meta-analysis (random effects model) of the odds-ratio for cardiovascular outcomes and breast cancer, between women with PCOS and controls. Funnel plots are not shown because less than 10 studies were included.

Cardiovascular morbidity and mortality

A cross-sectional report from the Northern Finland Birth Cohort estimated a 6.8% prevalence of CVD in women with PCOS by age 46 years—a figure double that of controls—with increased frequencies of myocardial infarction (1.8% versus 0.5%) and heart failure (0.7% versus 0.0%) in patients with PCOS (Ollila *et al.*, 2019). Two women with PCOS in this cohort had died from myocardial infarction compared to none in the control group (Ollila *et al.*, 2019). Another cross sectional report of patients in the Rancho Bernardo Study found a similar prevalence of CVD in women with PCOS and controls by age 70 years (27.3% versus 24.4%), but when restricting the analysis to women without diabetes or previous oophorectomy, there was in increase in CVD directly proportional to age and number of PCOS features

(Krentz et al., 2007). In a Czech report, coronary artery disease was also more prevalent in perimenopausal and menopausal women with PCOS (21%) compared with controls from the general population (5%), despite no differences being found between both groups in classic cardiovascular risk factors other than diabetes (Cibula et al., 2000). Both cardiovascular events and cardiovascular disease were more prevalent among menopausal mothers of women with PCOS who reported a history of PCOS symptoms compared with the mothers of women with PCOS who did not report such a history, even though myocardial infarction and stroke were similarly prevalent (Cheang et al., 2008). Regarding cerebrovascular disease, the frequencies of cerebral white matter lesions and silent cerebral infarcts in postmenopausal women with PCOS were double those found in controls with similar age and prevalence of hypertension, yet the patients were heavier and had higher LDL cholesterol levels (Guoqing et al., 2016).

However, other reports did not replicate these findings (Elting et al., 2001; Schmidt et al., 2011b; Polotsky et al., 2014; Merz et al., 2016; Meun et al., 2018, 2020; Forslund et al., 2022b). Unlike hypertension or diabetes, the prevalence of cardiac complaints was not increased in perimenopausal Dutch women with PCOS (Elting et al., 2001). Compared with controls, women with PCOS who submitted to ovarian wedge resection did not show an increased incidence of CVD or increased rates of cardiovascular death after 21 years of follow-up in one study (Schmidt et al., 2011b; Forslund et al., 2022b), even though hypertension, early onset of smoking and hypertriglyceridaemia were more frequent in patients (Schmidt et al., 2011b). Another Swedish series reported similar ORs for myocardial infarction and stroke in women with PCOS diagnosed according to NIH criteria 24 years earlier, compared with appropriate controls (Forslund et al., 2022a). Other reports that support these findings describe no increase in 10-year CVD risk (Meun et al., 2020) or cumulative 10-year cardiovascular mortality (Merz et al., 2016) in menopausal women with PCOS, nor in cardiovascular death after 12 years of follow-up of women with hyperandrogenaemia and irregular menses (Polotsky et al., 2014). Nevertheless, these negative findings could be related to a selection bias since, by including women in whom menopause occurred several years earlier, these studies might have missed women who may have already died from CVD before being eligible for inclusion (Meun et al., 2018).

Meta-analysis of available studies showed no increase in the OR for CVD (7 studies; 1293 women with PCOS; 5745 controls; OR 1.16 (0.75, 1.80); very low certainty of evidence; Fig. 8) and coronary artery disease (4 studies; 738 women with PCOS; 2729 controls; OR 1.38 (0.62, 3.10) very low certainty of evidence; Fig. 8) in peri- and postmenopausal women with PCOS.

On the contrary, the prevalence of myocardial infarction (3 studies; 338 women with PCOS; 1766 controls; OR 2.51 (1.08, 5.81); low certainty of evidence) and stroke (4 studies; 957 women with PCOS; 2826 controls; OR 1.75 (1.03, 2.99); very low certainty of evidence, Fig. 8) was increased in these patients compared with controls. However, in three of the four studies included in these meta-analyses women with PCOS showed increased BMI values and/or frequency of obesity compared with the controls (Wild et al., 2000; Ollila et al., 2016; Forslund et al., 2022a), and in the only study in which BMI was similar in women with PCOS and controls (Schmidt et al., 2011b), the prevalence of myocardial infarction and stroke did not differ among groups. Another study (Cheang et al., 2008) was not included in these meta-analyses because both groups of menopausal women, those with or without PCOS, were recruited among mothers of premenopausal women with PCOS, and this selection bias could have influenced the associations found. Other individual CVD outcomes, such as angor pectoris or transient ischaemic attack, were reported in less than three of these reports and did not undergo meta-analysis, yet these outcomes, as well as all-cause mortality, CVD-related mortality and CVD, were not more frequent in women with PCOS compared with controls in the study with the longer follow-up in which mean age of menopausal women was above 80 years (Forslund et al., 2022b).

Menopausal morbidity and gynaecological cancer

Climacteric symptoms

Climacteric symptoms were addressed in three reports (Dahlgren et al., 1992; Schmidt et al., 2011a; Yin et al., 2018). Only in one study were there differences in symptoms, such as sweating and hot flushes, which were more frequent in women with PCOS, or vaginal dryness, which was less frequent in controls (Schmidt et al., 2011a). No differences among groups were observed in vasomotor crisis or signs of oestrogen deficiency in other reports (Dahlgren et al., 1992; Yin et al., 2018).

Breast cancer

The risk for breast cancer was not increased in menopausal women with PCOS or polycystic ovaries compared with controls in six reports from five studies (Gammon and Thompson, 1991; Anderson et al., 1997; Parazzini et al., 1997; Schmidt et al., 2011b; Kim et al., 2016; Forslund et al., 2022b) although in one of them (Anderson et al., 1997), benign breast disease was more common in the former.

Meta-analysis of these studies showed that the OR for having breast cancer was not increased in women with PCOS compared with controls (5 studies; 618 women with PCOS; 44 233 controls; OR 1.06 (0.78, 1.43); low certainty of evidence; Fig. 8). The risk of breast cancer after menopause was increased by overweight, obesity, diabetes and nulliparity, regardless of PCOS status (Kim et al., 2016).

Endometrial cancer

In general, endometrial cancer risk was inversely associated with age at menarche and directly associated with age at menopause and years of menstruation (Zucchetto et al., 2009). Multiparity strongly reduced the risk among women under age 60 years (Zucchetto et al., 2009). Oral contraceptive use conferred a 40% reduced risk, irrespective of time since cessation (Zucchetto et al., 2009). Even though the association of PCOS with endometrial cancer in premenopausal women is clearly established (Iatrakis et al., 2006), and is explained by chronic oestrogen stimulation of the endometrium in the absence of periodical progesterone deprivation (Balen, 2001; Ding et al., 2018), no cases of this and other gynaecological neoplasia developed in 32 menopausal women and 95 controls after 21 years of follow-up in one series (Schmidt et al., 2011b).

Psychiatric disorders

Even though anxiety and depression are common during the menopausal transition regardless of PCOS (Mulhall et al., 2018), one report described that, as happens in younger women, the prevalence of these disorders was increased in women with PCOS by 46 years of age (Karjula et al., 2017), in association with reduced health-related quality of life, decreased life satisfaction, and a poorer self-reported health status compared with control women (Karjula et al., 2020). However, health-related quality of life was not different in peri- and postmenopausal women with

PCOS compared with similarly aged controls in another study (Forslund *et al.*, 2022a).

Bone mineral metabolism and disease

No differences have been found between menopausal women with PCOS and appropriate controls in bone mineral density, number of bone fractures, prevalence of osteoporosis, and use of osteoporosis-related drugs (Schmidt *et al.*, 2012). However, PCOS was associated with the development of knee osteoarthritis in postmenopausal women in a recent report (Deniz and Kehribar, 2021).

Self-reported data on symptoms, diagnosed diseases, and medication and healthcare service use

A recent study showed that, by age 46 years, women with PCOS had increased risk ratios for overall morbidity and medication use even after adjusting for BMI (Kujanpää *et al.*, 2022). In this study (Kujanpää *et al.*, 2022), diagnoses with increased prevalence in women with PCOS included migraine, hypertension, tendinitis, osteoarthritis, fractures, and endometriosis.

Sensitivity analyses

Diagnosis of PCOS

In some of the studies, particularly when the diagnosis of PCOS was assigned by the phenotype of the menopausal women, the criteria employed did not entirely match the standard NIH, Rotterdam or AE-PCOS definitions, and the use of other diagnostic criteria could influence the results. Hence, we restricted meta-analyses to studies applying these standard definitions or that used questionnaires that were validated against any of these definitions (Dahlgren *et al.*, 1992, 1994; Cibula *et al.*, 2000; Wild *et al.*, 2000; Alsamarai *et al.*, 2009; Puurunen *et al.*, 2009, 2011; Markopoulos *et al.*, 2011, 2013; Schmidt *et al.*, 2011a,b, 2017; Chan *et al.*, 2013; Snyder *et al.*, 2014; Echiburu *et al.*, 2016; Guoqing *et al.*, 2016; Ollila *et al.*, 2016, 2017, 2019; Forslund *et al.*, 2019, 2020, 2021, 2022a,b; Meun *et al.*, 2020; Persson *et al.*, 2023).

These meta-analyses are summarized in Table 4 and their forest plots are shown in Supplementary Fig. S3a and b. Peri- and postmenopausal women with PCOS presented, when compared with controls, increased BMI, waist circumference, serum androgens, fasting insulin and glucose concentrations and TG, and reduced SHBG and HDL-cholesterol levels, together with increased SBP values and increased ORs for diabetes and hypertension. No differences in age at menopause, waist-to-hip ratio, in other lipids, in DBP values or in the ORs for CVD or stroke were found. These results suggested a minor impact, if any, of the criteria used in the different studies for the definition of PCOS on the outcomes studied here. As before, one study was left out of the meta-analyses of hypertension and blood pressure values because women with PCOS and controls were matched for prevalence of hypertension (Guoqing *et al.*, 2016).

Population-based or non-population-based studies

Referral bias may exert a major impact on the presentation of PCOS, explaining some differences between patients recruited at clinical practices compared with those recruited from the general unselected population (Luque-Ramirez *et al.*, 2015). Accordingly, to avoid such a bias, we conducted a separate analysis restricted to the available population-based studies (Gammon and Thompson, 1991; Anderson *et al.*, 1997; Parazzini *et al.*, 1997; Margolin *et al.*, 2005; Krentz *et al.*, 2007; Gabrielli and Aquino, 2012; Polotsky *et al.*, 2012, 2014; Kim *et al.*, 2016; Ollila *et al.*, 2016, 2017, 2019; Meun *et al.*, 2018, 2020).

These meta-analyses are summarized in Table 5 and their forest plots are shown in Supplementary Fig. S4a and b. Peri- and postmenopausal women with PCOS presented, when compared with controls, increased BMI, waist circumference, serum total and free testosterone, fasting insulin and glucose concentrations, HOMA-IR values and TG, and reduced SHBG and HDL-cholesterol

Table 4. Meta-analysis of studies that compared peri- and postmenopausal women with PCOS with controls provided that the premenopausal diagnosis of PCOS met standard definitions (National Institutes of Health, Rotterdam, Androgen Excess and PCOS Society) or used questionnaires validated against standard PCOS criteria.

Outcome	Studies (n)	PCOS (n)	Controls (n)	SMD/OR	[95% CI]	Cohen's *U3*
Age at menopause	3	80	914	SMD 0.39	[−0.44. 1.22]	+15%
BMI	13	996	3204	**SMD 0.55**	**[0.33, 0.77]**	**+22%**
Waist circumference	8	704	2798	**SMD 0.47**	**[0.22, 0.72]**	**+19%**
Waist-to-hip ratio	9	478	1590	SMD 0.38	[0.00, 0.77]	+15%
Total testosterone	10	717	2044	**SMD 0.47**	**[0.30, 0.63]**	**+19%**
Free androgen index	7	620	1702	**SMD 0.98**	**[0.54, 1.41]**	**+38%**
Sex hormone-binding globulin	9	697	2011	**SMD −0.52**	**[−0.79, −0.26]**	**−21%**
Androstenedione	5	328	436	**SMD 0.82**	**[0.06, 1.59]**	**+32%**
Dehydroepiandrosterone-sulphate	4	116	228	**SMD 0.37**	**[0.01, 0.73]**	**+15%**
Homeostasis model assessment of insulin resistance	10	623	1849	**SMD 0.35**	**[0.17, 0.53]**	**+14%**
Fasting insulin	10	690	1916	**SMD 0.44**	**[0.23, 0.64]**	**+17%**
Fasting glucose	11	713	2616	**SMD 0.36**	**[0.20, 0.52]**	**+14%**
Diabetes	7	813	3437	**OR 3.21**	**[2.12, 4.88]**	
Total cholesterol	8	557	1468	SMD 0.05	[−0.10, 0.21]	+2%
High density lipoprotein-cholesterol	8	557	1468	**SMD −0.20**	**[−0.35, −0.04]**	**−8%**
Low density lipoprotein-cholesterol	7	537	1420	SMD 0.09	[−0.08, 0.26]	+4%
Triglycerides	8	557	1468	**SMD 0.21**	**[0.03, 0.39]**	**+8%**
Systolic blood pressure	8	724	2791	**SMD 0.21**	**[0.12, 0.30]**	**+8%**
Diastolic blood pressure	6	652	2680	SMD 0.09	[−0.09, 0.27]	+4%
Hypertension	7	844	3608	**OR 1.90**	**[1.33, 2.73]**	
Cardiovascular disease	4	1130	2932	OR 1.35	[0.72, 2.53]	
Myocardial infarction	3	338	1766	**OR 2.51**	**[1.08, 5.81]**	
Stroke	4	957	2826	**OR 1.75**	**[1.03, 2.99]**	

SMD, standardized mean difference; OR, odds ratio.
Data highlighted in boldface denote statistically significant effects. Forest plots are shown in Supplementary Fig. S3a and b.
Data on dyslipidaemia, coronary artery disease, and breast cancer were available in less than three reports, and were not meta-analysed.

Table 5. Meta-analysis of population-based studies that compared peri- and postmenopausal women with PCOS with controls.

Outcome	Studies (n)	PCOS (n)	Controls (n)	SMD/OR	[95% CI]	Cohen's *U3*
Age at menopause	3	278	24 197	SMD −0.09	[−0.41, 0.22]	−4%
BMI	9	1350	42 535	**SMD 0.46**	**[0.22, 0.69]**	**+18%**
Waist circumference	6	705	5004	**SMD 0.72**	**[0.45, 1.00]**	**+28%**
Waist-to-hip ratio	4	856	35 844	SMD 0.43	[−0.03, 0.90]	+17%
Total testosterone	7	703	4402	**SMD 1.19**	**[0.31, 2.07]**	**+45%**
Free androgen index	6	661	4206	**SMD 1.68**	**[0.99, 2.37]**	**+60%**
Sex hormone-binding globulin	7	727	4853	**SMD −0.49**	**[−0.65, −0.33]**	**−19%**
Androstenedione	4	379	1115	SMD 0.27	[−0.18, 0.72]	+11%
Homeostasis model assessment of insulin resistance	5	614	4585	**SMD 0.71**	**[0.25, 1.17]**	**+28%**
Fasting insulin	5	614	4585	**SMD 0.53**	**[0.21, 0.85]**	**+21%**
Fasting glucose	6	720	4756	**SMD 0.59**	**[0.31, 0.87]**	**+23%**
Diabetes	5	565	4635	**OR 2.45**	**[1.22, 4.90]**	
Total cholesterol	4	450	2128	SMD −0.05	[−0.27, 0.17]	−2%
High density lipoprotein-cholesterol	4	450	2128	**SMD −0.28**	**[−0.51, −0.05]**	**−11%**
Low density lipoprotein-cholesterol	3	344	1957	SMD 0.06	[−0.21, 0.33]	+2%
Triglycerides	4	450	2128	**SMD 0.34**	**[0.02, 0.66]**	**+12%**
Dyslipidaemia	3	180	3265	OR 1.28	[0.41, 3.97]	
Systolic blood pressure	5	687	3591	SMD 0.11	[−0.08, 0.30]	+4%
Diastolic blood pressure	5	687	3591	SMD 0.09	[−0.07, 0.25]	+4%
Hypertension	5	598	6109	**OR 1.65**	**[1.16, 2.35]**	
Cardiovascular disease	3	585	1948	**OR 1.69**	**[1.06, 2.69]**	
Breast cancer	4	586	44 138	OR 1.02	[0.72, 1.46]	

SMD, standardized mean difference; OR, odds ratio.
Data highlighted in boldface denote statistically significant effects. Forest plots are shown in Supplementary Fig. S4a and b.
Data on dehydroepiandrosterone-sulphate, coronary artery disease, stroke, and myocardial infarction were available in less than three reports, and were not meta-analysed.

levels, together with increased and increased ORs for diabetes, hypertension, and CVD. No differences in age at menopause, waist-to-hip ratio, androstenedione levels, in other lipids, in systolic and DBP values or in the ORs for breast cancer were found. The most remarkable finding of this sensitivity analysis consisted of the OR for CVD being increased in women with PCOS; however, such outcome was present only in three population-based studies and must be considered with caution. Overall, most results derived from population-based studies were similar to those of the general unrestricted meta-analysis.

Menopause

It was possible that the findings in perimenopausal women could differ from women with established menopause. For this reason, we restricted analyses to reports that included only postmenopausal women, with and without PCOS (Gammon and Thompson, 1991; Anderson *et al.*, 1997; Parazzini *et al.*, 1997; Margolin *et al.*, 2005; Krentz *et al.*, 2007; Puurunen *et al.*, 2011; Schmidt *et al.*, 2011a,b; Armeni *et al.*, 2013; Markopoulos *et al.*, 2013; Pinola *et al.*, 2015; Guoqing *et al.*, 2016; Merz *et al.*, 2016).

These meta-analyses are summarized in Table 6 and their forest plots are shown in Supplementary Fig. S5a and b. Postmenopausal women with PCOS presented, when compared with controls, increased BMI, waist circumference, waist-to-hip ratio, serum androgens and fasting insulin concentrations, and reduced SHBG and HDL-cholesterol levels. No differences in age at menopause, other surrogate indexes of insulin resistance, in fasting glucose, in other lipids, in blood pressure values or in the ORs for diabetes or hypertension were found. Hence, it appears that the hyperandrogenic phenotype and the association with obesity of women with PCOS persisted after menopause, yet the metabolic phenotype and the association with increased blood pressure was ameliorated substantially. Once again, one study was left out of the meta-analyses of hypertension and blood pressure values because women with PCOS and controls were matched for prevalence of hypertension (Guoqing *et al.*, 2016).

Ovarian surgery

Ovarian surgery—including ovarian wedge resection—could impact the outcomes studied here by avoiding or ameliorating the long-term exposure to androgen excess in the subset of patients with PCOS submitted to these surgical procedures (Hendriks *et al.*, 2007). Hence, we repeated meta-analyses after excluding studies that reported data from women with a history of surgical menopause or ovarian wedge resection (Gammon and Thompson, 1991; Dahlgren *et al.*, 1992, 1994; Anderson *et al.*, 1997; Cibula *et al.*, 2000; Talbott *et al.*, 2000, 2008; Wild *et al.*, 2000; Margolin *et al.*, 2005; Schmidt *et al.*, 2011a,b; 2017; Forslund *et al.*, 2021, 2022b).

These meta-analyses showed that peri- and postmenopausal women with PCOS, when compared with controls, showed increased serum TT and androstenedione concentrations and FAI, decreased SHBG, increased BMI, waist circumference, waist-to-hip ratio, HOMA-IR, and fasting insulin and glucose levels, decreased HDL-cholesterol and increased TG and SBP, as well as increased ORs for diabetes and hypertension, confirming almost all the results of the original meta-analyses (Table 7 and Supplementary Fig. S6a and b). Therefore, results of this sensitivity analysis replicated those of the general meta-analyses. Noteworthy, is that a recent study found no differences in cardiovascular risk factors, such as hypertension or diabetes, among perimenopausal women with PCOS with a history of ovarian wedge resection compared with similarly aged patients without previous ovarian surgery (Forslund *et al.*, 2023).

Weight excess

Meta-analysis of BMI showed increased values in the peri- and postmenopausal women with PCOS compared with controls included in the meta-analyses of other outcomes. This mismatch could influence the serum androgen concentrations and cardiometabolic associations of PCOS described earlier. Hence, we repeated such meta-analyses selecting only reports in which the BMI was similar in women with PCOS and controls (Dahlgren *et al.*, 1992, 1994; Anderson *et al.*, 1997; Cibula *et al.*, 2000; Winters

Table 6. Meta-analysis in studies that compared postmenopausal women with PCOS with similarly aged controls.

Outcome	Studies (n)	PCOS (n)	Controls (n)	SMD/OR	[95CI]	Cohen's *U3*
Age at menopause	3	168	23 705	SMD −0.32	[−0.69, 0.06]	−13%
BMI	11	767	36 080	**SMD 0.74**	**[0.29, 1.19]**	**+29%**
Waist circumference	7	191	1349	**SMD 0.77**	**[0.25, 1.29]**	**+30%**
Waist-to-hip ratio	7	671	35 304	**SMD 0.37**	**[0.09, 0.65]**	**+15%**
Total testosterone	9	270	1447	**SMD 0.47**	**[0.25, 0.69]**	**+19%**
Free androgen index	7	179	554	**SMD 1.21**	**[0.62, 1.80]**	**+45%**
Sex hormone-binding globulin	9	270	1447	**SMD −0.76**	**[−1.00, −0.51]**	**−30%**
Androstenedione	7	200	1129	**SMD 0.63**	**[0.09, 1.16]**	**+25%**
Dehydroepiandrosterone-sulphate	6	188	1032	**SMD 0.38**	**[0.03, 0.73]**	**+15%**
Homeostasis model assessment of insulin resistance	8	265	1410	SMD 0.55	[0.00, 1.10]	+22%
Fasting insulin	6	170	1000	**SMD 0.87**	**[0.38, 1.36]**	**+34%**
Fasting glucose	7	195	1270	SMD 0.54	[−0.00, 1.09]	+21%
Diabetes	4	78	480	OR 2.32	[0.63, 8.50]	
Total cholesterol	5	229	1368	SMD 0.10	[−0.09, 0.29]	+4%
High density lipoprotein-cholesterol	5	229	1368	**SMD −0.45**	**[−0.67, −0.23]**	**−18%**
Low density lipoprotein-cholesterol	5	229	1368	SMD 0.15	[−0.09, 0.39]	+6%
Triglycerides	5	229	1368	**SMD 0.49**	**[0.24, 0.74]**	**+19%**
Systolic blood pressure	4	159	1228	SMD 0.15	[−0.19, 0.49]	+5%
Diastolic blood pressure	4	159	1228	SMD 0.18	[−0.13, 0.49]	+7%
Hypertension	4	107	705	OR 2.04	[0.67, 6.20]	
Breast cancer	4	551	41 236	OR 0.93	[0.63, 1.36]	

SMD, standardized mean difference; OR, odds ratio.
Data highlighted in boldface denote statistically significant effects. Forest plots are shown in Supplementary Fig. S5a and b.
Data on dyslipidaemia, cardiovascular disease, coronary artery disease, and stroke were available in less than three reports and were not meta-analysed.

Table 7. Meta-analysis in studies that compared peri- and postmenopausal women with PCOS with controls after excluding studies including women submitted to wedge resection or ovarian surgery.

Outcome	Studies (n)	PCOS (n)	Controls (n)	SMD/OR	[95% CI]	Cohen's *U3*
Age at menopause	4	212	1194	SMD 0.07	[−0.61, 0.76]	+3%
BMI	19	1341	1537	**SMD 0.63**	**[0.43, 0.82]**	**+25%**
Waist circumference	11	980	5735	**SMD 0.67**	**[0.43, 0.91]**	**+26%**
Waist-to-hip ratio	11	677	2386	**SMD 0.49**	**[0.25, 0.73]**	**+19%**
Total testosterone	15	1175	5095	**SMD 0.78**	**[0.29, 1.28]**	**+30%**
Free androgen index	12	1061	4803	**SMD 1.28**	**[0.84, 1.71]**	**+48%**
Sex hormone-binding globulin	14	1179	5513	**SMD −0.56**	**[−0.74, −0.39]**	**−22%**
Androstenedione	8	610	1467	**SMD 0.52**	**[0.12, 0.91]**	**+21%**
Dehydroepiandrosterone-sulphate	6	292	1088	SMD 0.18	[−0.02, 0.39]	+7%
Homeostasis model assessment of insulin resistance	14	927	5538	**SMD 0.56**	**[0.26, 0.86]**	**+22%**
Fasting insulin	13	964	5283	**SMD 0.61**	**[0.38, 0.85]**	**+24%**
Fasting glucose	15	1095	5724	**SMD 0.51**	**[0.31, 0.72]**	**+20%**
Diabetes	8	663	7303	**OR 2.11**	**[1.17, 3.82]**	
Total cholesterol	11	822	3089	SMD −0.02	[−0.12, 0.16]	+1%
High-density lipoprotein-cholesterol	11	822	3089	**SMD −0.33**	**[−0.49, −0.18]**	**−13%**
Low-density lipoprotein-cholesterol	9	696	2870	SMD 0.08	[−0.07, 0.24]	+3%
Triglycerides	11	822	3089	**SMD 0.33**	**[0.17, 0.48]**	**+13%**
Dyslipidaemia	3	198	3438	OR 0.99	[0.55, 1.76]	
Systolic blood pressure	11	989	4412	**SMD 0.14**	**[0.01, 0.27]**	**+6%**
Diastolic blood pressure	11	989	4412	SMD 0.04	[−0.10, 0.17]	+2%
Hypertension	9	748	9033	**OR 1.78**	**[1.22, 2.60]**	
Cardiovascular disease	5	642	4590	OR 1.26	[0.74, 2.15]	

SMD, standardized mean difference; OR, odds ratio.
Data highlighted in boldface denote statistically significant effects. Forest plots are shown in Supplementary Fig. S6a and b.
Data on coronary artery disease, stroke, myocardial infarction, and breast cancer were available in less than three reports and were not meta-analysed.

et al., 2000; Elting *et al.*, 2001; Schmidt *et al.*, 2011b, 2017; Markopoulos *et al.*, 2013; Echiburu *et al.*, 2016; Kim *et al.*, 2016; Merz *et al.*, 2016; Forslund *et al.*, 2021, 2022b).

These meta-analyses are summarized in Table 8 and their forest plots are shown in Supplementary Fig. S7a and b. The only difference found after excluding reports mismatched for BMI consisted of a decrease in HDL-cholesterol concentrations in women with PCOS, compared with controls. The differences among groups found when considering all available reports as a whole regarding BMI, waist circumference, waist-to-hip ratio, serum androgens and SHBG levels, HOMA-IR, fasting insulin and glucose, TG, SBP, and in the ORs for diabetes and hypertension, were no longer found when restricting the analyses to reports in which BMI was similar among women with PCOS and controls. Of note was that TT, FAI, SHBG, androstenedione and DHEAS concentrations were available in less than three reports and were not meta-analysed.

Serum androgen concentrations were reported in only two of the studies in which patient and controls showed similar BMI—the Gothenburg's cohort (Dahlgren *et al.*, 1992, 1994; Schmidt *et al.*, 2011a,b; 2017; Forslund *et al.*, 2021, 2022b) and Markopoulos *et al.* (2011, 2013); of note, these studies included only postmenopausal women. Even though we did not perform meta-analysis of these two studies, in both of them menopausal women with PCOS

Table 8. Meta-analysis in studies that compared peri- and postmenopausal women with PCOS with controls who were similar in terms of BMI.

Outcome	Studies (n)	PCOS (n)	Controls (n)	SMD/OR	[95% CI]	Cohen's *U3*
Age at menopause	4	196	24 457	SMD −0.19	[−0.51, 0.13]	−8%
BMI	7	672	38 694	SMD 0.04	[−0.04, 0.11]	+2%
Waist circumference	5	133	1156	SMD 0.15	[−0.05, 0.34]	+6%
Waist-to-hip ratio	6	605	35 697	SMD 0.02	[−0.14, 0.17]	+1%
Homeostasis model assessment of insulin resistance	4	105	404	SMD 0.02	[−0.34, 0.39]	+1%
Fasting insulin	3	80	134	SMD 0.19	[−0.28, 0.66]	+8%
Fasting glucose	5	133	1156	SMD 0.13	[−0.07, 0.32]	+5%
Diabetes	5	187	6486	OR 1.71	[0.64, 4.55]	
Total cholesterol	4	108	1132	SMD −0.03	[−0.24, 0.18]	−1%
High-density lipoprotein-cholesterol	4	108	1132	**SMD −0.29**	**[−0.56, −0.02]**	**−12%**
Low-density lipoprotein-cholesterol	4	108	1132	SMD −0.06	[−0.27, 0.15]	−2%
Triglycerides	4	108	1132	SMD 0.24	[−0.12, 0.60]	+5%
Dyslipidemia	3	113	3322	OR 1.05	[0.59, 1.85]	
Systolic blood pressure	4	108	1132	SMD 0.10	[−0.12, 0.31]	+4%
Diastolic blood pressure	4	108	1132	SMD −0.08	[−0.41, 0.25]	−3%
Hypertension	6	214	6528	OR 1.41	[0.79, 2.50]	
Breast cancer	3	571	37 455	OR 1.25	[0.88, 1.77]	

SMD, standardized mean difference; OR, odds ratio.
Data highlighted in boldface denote statistically significant effects. Forest plots are shown in Supplementary Fig. S7a and b.
Data on total testosterone, free androgen index, sex hormone-binding globulin, androstenedione, dehydroepiandrosterone-sulphate, cardiovascular disease, coronary artery disease, myocardial infarction, and stroke were available in less than three reports, and were not meta-analysed.

presented with increased serum TT, FAI, androstenedione and DHEAS, and reduced SHBG concentrations (Dahlgren et al., 1992; Markopoulos et al., 2013), suggesting that the hyperandrogenic phenotype of PCOS actually persists after menopause regardless of obesity. Importantly, one of the latest reports of the Gothenburg's cohort showed that, when these women reached a mean age above 80 years, differences in serum androgen levels were no longer statistically significant, even though hirsutism was still much more prevent in elderly women with PCOS compared with appropriate controls (Forslund et al., 2021).

Unfortunately, the paucity of studies restricted to postmenopausal patients and controls presenting with similar BMI precluded appropriate meta-analyses. On an individual basis, the few available studies showed that, when compared with control women, menopausal women with PCOS presented with reduced fertility and increased frequencies of benign breast disease and surgical menopause (Anderson et al., 1997), higher androgens levels, insulin secretion indices, and high sensitivity C-reactive protein levels, with no differences in fasting glucose, insulin and adipocytokines levels, and indexes of insulin sensitivity and resistance (Markopoulos et al., 2013), increased androgens and self-reported hirsutism but reduced SHBG and FSH levels (Dahlgren et al., 1992; Schmidt et al., 2011a; Forslund et al., 2021), higher TG and a larger prevalence of treated hypertension without differences in cardiovascular events and prevalence of diabetes, cancer, and mortality (Schmidt et al., 2011b) and a reduced prevalence of hypothyroidism (Schmidt et al., 2017).

Management of PCOS during the menopausal transition

There is no consensus at present on how to diagnose PCOS in menopausal women. Most studies assigned the diagnosis of PCOS retrospectively, based on earlier clinical history, or were follow-up studies of women that were diagnosed during their late fertile years using NIH or ESHRE/ASRM criteria, or even by histopathology after ovarian wedge resection (Cibula et al., 2000) or ovariectomy (Dahlgren et al., 1992, 1994).

Several authors proposed specific criteria for the diagnosis of PCOS in menopausal women, summarized in Table 9 (Krentz et al., 2007; Armeni et al., 2013; Gabrielli et al., 2015). It is noteworthy that, in addition to androgen excess and a history of oligo-ovulation or non-male factor infertility, these definitions include metabolic variables, such as indexes of insulin resistance or obesity, that are not considered by currently accepted definitions of PCOS in premenopausal women. The use of these novel definitions that include metabolic criteria may have predetermined the association of PCOS with metabolic derangements reported in these studies (Krentz et al., 2007; Armeni et al., 2013; Gabrielli et al., 2015).

Finally, we have not found any data specifically addressing treatment of PCOS in women over age 45 years.

Discussion

Our systematic review and meta-analysis confirms that most of the clinical and biochemical characteristics of premenopausal women with PCOS, as well as many of its cardiometabolic associations (Escobar-Morreale, 2018), persist during their late-reproductive years and after menopause. Overall, when compared with similarly aged controls, women with PCOS presented with increased serum androgen levels and decreased SHBG concentrations, increased BMI, waist circumference and waist-to-hip ratios, increased surrogate indexes of insulin resistance and increased fasting glucose concentrations, decreased HDL-cholesterol and increased TG, and SBP values. Accordingly, the ORs for diabetes, hypertension, myocardial infarction and stroke were also increased in patients with PCOS yet, importantly, the ORs for dyslipidaemia, CVD as a whole, coronary artery disease, and breast cancer were similar to those of controls.

Our results regarding cardiovascular outcomes were in agreement with the long-known paradox that there is no definite proof that women with PCOS have more cardiovascular events than women from the general population (Legro, 2003) even though these women may show almost all classic and non-classic cardiovascular risk factors (Wild et al., 2010). The issue was summarized by the latest guidelines for the management of PCOS (Teede et al., 2018) that included a clinical practice point stating that 'cardiovascular disease risk in women with PCOS remains unclear pending high-quality studies; however prevalence of cardiovascular

Table 9. Definitions and criteria proposed for the diagnosis of PCOS in menopausal women*.

Armeni *et al.*, 2013	Gabrielli *et al.*, 2015	Krentz *et al.*, 2007
Prevalence of PCOS in the study: 15% History of clinical hyperandrogenism: • Hirsutism. • Alopecia. and/or Current biochemical androgen excess: • Free testosterone in the highest quintile. • SHBG in the lowest quintile.	**Prevalence of PCOS in the study: 7.6%** Clinical hyperandrogenism: • Score >5 in a hirsutism score validated for menopausal women and/or Current biochemical androgen excess: • Total testosterone >95th percentile.	**Prevalence of PCOS in the study: 9.3%** Current or correctly documented earlier clinical and/or biochemical hyperandrogenism.
History of menstrual irregularity	History of oligomenorrhoea	History of oligomenorrhoea, infertility or abortion.
Insulin resistance: • HOMA-IR in the highest quintile. • Fasting glycaemia in the highest quintile.	Insulin resistance: • HOMA-IR >75th percentile.	Insulin resistance: • HOMA-IR in the highest quartile. • Fasting glycaemia>6.1 mmol/L
Central obesity Non-male factor infertility		Central obesity

* All required ≥3 criteria to fulfil the diagnosis of PCOS.
HOMA-IR, homeostasis model assessment of insulin resistance; SHBG, sex hormone-binding globulin.

disease risk factors is increased, warranting consideration of screening'.

Our sensitivity analyses, however, provided further insights into these findings. Neither a history of ovarian surgery or wedge resection nor the differences in criteria used to diagnose PCOS or the setting—referral or population-based—where the studies were conducted appeared to exert any important influence on the results of the general meta-analyses. On the contrary, when analyses were restricted to studies that only considered postmenopausal women, only increased BMI, indexes of abdominal adiposity, serum androgens and fasting insulin concentrations, and reduced SHBG and HDL-cholesterol levels remained. Hence, these data indicate that some of the most severe cardiometabolic associations of PCOS—including diabetes or hypertension—may not persist after menopause; whether this happens as the result of an amelioration of PCOS after the cessation of normal ovarian function, or translates to a worsening of metabolic function in control women after menopause, remains to be elucidated (Welt and Carmina, 2013).

Moreover, when we restricted meta-analyses to reports in which patients with PCOS and controls were selected so as to have similar BMI—including both peri- and postmenopausal women as a whole—all differences in cardiometabolic outcomes observed earlier lost statistical significance, with the exception of reduced HDL-cholesterol concentrations. In only two studies, postmenopausal women with or without PCOS showed similar mean BMIs, and in these studies serum androgen concentrations were increased and SHBG concentrations were reduced in the former, suggesting that persistence of androgen excess after menopause was the main finding in these women. Moreover, even the increase in the frequencies of myocardial infarction and stroke found by our meta-analyses may have been influenced by the association of PCOS with weight excess: such analyses included four studies, of which three showed increased BMI and/or frequency of obesity in women with PCOS compared with controls (Wild *et al.*, 2000; Ollila *et al.*, 2016; Forslund *et al.*, 2022a). Furthermore, in the other study that included women with PCOS and controls presenting with similar mean BMI values, the frequencies of these cardiovascular events were not different among groups (Schmidt *et al.*, 2011b).

Overall, these findings are in agreement with those of earlier meta-analyses concluding that, even though PCOS may be associated with hypertension, diabetes, dyslipidaemia, and CVD before menopause (Amiri *et al.*, 2020; Ramezani Tehrani *et al.*, 2020; Wekker *et al.*, 2020), having a history of PCOS during reproductive years may not be an important factor for developing hypertension and/or cardiovascular events later in life (Amiri *et al.*, 2020; Ramezani Tehrani *et al.*, 2020). Taken as a whole, available evidence supports that, even though hyperandrogenism may persist for years after menopause, the PCOS phenotype ameliorates with ageing and that there is no further deterioration in cardiometabolic profile in women with PCOS after menopause (Helvaci and Yildiz, 2022). A non-mutually exclusive alternative explanation resides on the fact that, unlike unaffected women, a considerable number of women with PCOS included in these studies may have been on treatment for PCOS and associated cardiometabolic comorbidities before menopause, preventing the development of these complications.

In other words, weight excess and abdominal adiposity appear to play a major role in almost all the cardiometabolic associations of PCOS, also in late reproductive years and menopause—as also happens in younger patients (Luque-Ramirez *et al.*, 2007; Hernández-Mijares *et al.*, 2013; Anagnostis *et al.*, 2021; Falcetta *et al.*, 2021)—to the extent that only a relatively mild clinical and biochemical hyperandrogenism persists after the physiological cessation of ovarian function. Whether this mild androgen excess of menopausal women with PCOS originates at the adrenals or arises from residual ovarian theca cell function under the influence of increased LH levels is unclear at present (Puurunen *et al.*, 2009, 2011; Markopoulos *et al.*, 2011).

Aside from the studies meta-analyzed here, the lack of association of hyperandrogenism with metabolic dysfunction in menopausal women is also supported by a secondary analysis of the women included in the Diabetes Prevention Program and the Diabetes Prevention/Program Outcomes Study (Kim *et al.*, 2018), in which neither the FAI nor menstrual irregularity were associated with the risk of developing type 2 diabetes or presenting with coronary artery calcification. Moreover, the fact that abdominal obesity and insulin resistance were used as criteria for the diagnosis of PCOS after menopause in a few studies (Krentz *et al.*, 2007; Armeni *et al.*, 2013; Gabrielli *et al.*, 2015) may also facilitate the association of PCOS with cardiometabolic risk factors in menopausal women by a selection bias.

In summary, our meta-analyses indicated that postmenopausal PCOS was characterized by androgen excess, with cardiometabolic associations being determined mostly by the presence of weight excess and obesity, and an overall cardiovascular risk that was similar to that of postmenopausal women without the syndrome and lower than previously anticipated based on their risk during reproductive years (Helvaci and Yildiz, 2022). Noteworthy, a recent study indicated that insulin resistance, androgens, and lipids in lean women with PCOS gradually improved from early to late-reproductive age, suggesting that by avoiding obesity their cardiometabolic risk would be attenuated (Livadas *et al.*, 2020). The obvious clinical relevance of these findings includes the reassurance that having a diagnosis of PCOS does not necessarily worsen a woman's long-term cardiometabolic and cancer prognosis provided that obesity is avoided, and the need for prescribing strategies directed towards prevention and treatment of weight excess as early as possible in affected women.

Strengths and limitations

The strengths of our study derive mostly from the large number of studies, patients, and controls finally included in the meta-analyses. However, there are weaknesses that might limit the generalization of our analyses and conclusions, starting with the very low to moderate quality of the evidence of the outcomes analyzed here. Several factors reduced such quality, but the most important were limitations of the original studies derived from the risk of bias from unexplained heterogeneity between studies in estimates of the diverse associations, imprecision of results, and residual confounding. Also, studies matched both for BMI and for pre- or postmenopausal status were too few to actually meta-analyse some of their outcomes, as was our intention. Moreover, the decrease in the numbers of women with PCOS and controls being included in the sensitivity studies may have contributed to the loss of the statistically significant differences found in some outcomes when all reports were included in such meta-analyses as a whole. In addition, many if not most studies used cross-sectional case–control designs in which the diagnosis of PCOS in menopausal women was assigned retrospectively, from epidemiological or clinical reports in some cases, and occasionally were even based on self-report by the participants; unfortunately, prospective long-term follow-up studies that applied a rigorous physician-based assignment of the diagnosis of PCOS were scarce, and firm conclusions on the issue would definitely need further studies of this kind.

Moreover, there was substantial variability in the definitions used to assign the diagnosis of PCOS retrospectively, even though we have taken into account such heterogeneity and applied a random-effects model in all meta-analyses for this reason. Our sensitivity analysis, on the contrary, demonstrated only a minor impact of this variability in criteria on the final results.

Furthermore, in the few studies that assigned the diagnosis of PCOS prospectively to menopausal women, there was no standardization or consensus at all on the criteria to be applied in that setting. Obviously, menstrual irregularity or oligo-ovulation cannot be used as a criterion in menopausal women. Yet also, the few proposals made on the issue (Krentz *et al.*, 2007; Armeni *et al.*, 2013; Gabrielli *et al.*, 2015) included metabolic variables that are not currently considered among the criteria of PCOS used for their premenopausal counterparts, biasing results towards an association of PCOS with metabolic derangements. To further complicate the issue, the methods used for the assessment of clinical and biochemical androgen excess and ovarian morphology have not been validated for menopausal women, and reference values to discriminate between women with PCOS and women who develop mild symptoms of androgen excess as a consequence of the physiological decline in oestrogens that follows menopause are definitely lacking.

Noteworthy is that, albeit PCOS appears to be the most frequent androgen excess disorder not only in premenopausal women but also after menopause (Elhassan *et al.*, 2018), ovarian hyperthecosis and androgen secreting tumours cause androgen excess more frequently in menopausal women than in their premenopausal counterparts (Elhassan *et al.*, 2018) and must be considered in the differential diagnosis of all menopausal women with symptoms of androgen excess. Importantly, cut-off values of serum androgen concentrations have not been validated for the differential diagnosis between PCOS and non-functional causes of androgen excess in menopausal women. Until the issue is addressed in future studies, we suggest to rely mostly on clinical presentation for such differential diagnosis, imaging immediately the ovaries and the adrenals of women who present with a sudden onset of rapidly progressing and severe clinical hyperandrogenism, particularly when associated with signs of masculinization or defeminization (Alpanes *et al.*, 2012).

Future directions for clinical research

According to our systematic review and meta-analyses of the consequences of PCOS during the menopausal transition and afterwards, a substantial improvement in the quality of the evidence currently available is definitely needed. In this regard, priority should be given to conducting large multicentre prospective studies that monitor women with PCOS and similarly aged controls during the menopausal transition and after menopause. Ideally, these studies should include patients presenting with all phenotypes of PCOS, as well as control women in whom such a diagnosis has been excluded with certainty. To avoid selection bias, these subjects should be recruited from the general population and not from clinical practices, whenever possible. Also, the diagnosis of PCOS should be assigned before menopause and should be confirmed at recruitment using widely accepted definitions and criteria for the syndrome. Particular attention must be given to the presence of weight excess, considering the major impact that obesity exerted on the cardiometabolic associations of PCOS according to our present results. And, finally, confounding factors, such as ethnicity, body composition, and medical interventions, must be recorded and considered for the final analysis of the outcomes.

Conclusion

Clinical and biochemical hyperandrogenism appeared to persist during the late-reproductive years and after menopause in women with PCOS. In contrast, most cardiometabolic comorbidities, such as insulin resistance, diabetes, atherogenic dyslipidaemia, hypertension, myocardial infarction, and stroke, were actually related to the frequent coexistence of weight excess with PCOS, indicating the importance of targeting obesity in this population as early as possible. Moreover, cardiovascular events and breast cancer were not more frequent in women with PCOS. However, the overall low quality of the evidence, methodological differences, and dissimilar results highlight the need for conducting adequately powered prospective studies in order to improve current knowledge on the menopausal transition in women with

PCOS on which to base management guidelines for these women, as no such guidelines exist at present.

Supplementary data

Supplementary data are available at *Human Reproduction Update* online.

Data availability

The data underlying this article are available in the article and in its online supplementary material.

Authors' roles

H.F.E.-M. conceived the original idea. M.M.-d.M., M.L.-R., and H.F.E.-M. developed the study protocol. M.M.-d.M., M.L.-R., and L.N.-C. performed study selection and data extraction. M.M.-d.M. and H.F.E.-M. performed the statistical analysis. M.L.-R. and L.N.-C. assessed quality of the studies. M.M.-d.M. wrote the original draft and H.F.E.-M wrote the final version of the manuscript. M.L.-R. and L.N.-C. reviewed and edited the manuscript and contributed intellectually to its content. All authors supervised the findings of this work.

Funding

This work was supported by Instituto de Salud Carlos III (grant numbers PI18/01122 and PI21/00116), and cofinanced by the European Development Regional Fund 'A way to achieve Europe' (EDRF). CIBERDEM and IRYCIS are also initiatives of the Instituto de Salud Carlos III. The funding sources of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

Conflict of interest

We declare no competing interests.

References

Alpanes M, Gonzalez-Casbas JM, Sanchez J, Pian H, Escobar-Morreale HF. Management of postmenopausal virilization. *J Clin Endocrinol Metab* 2012;**97**:2584–2588.

Alsamarai S, Adams JM, Murphy MK, Post MD, Hayden DL, Hall JE, Welt CK. Criteria for polycystic ovarian morphology in polycystic ovary syndrome as a function of age. *J Clin Endocrinol Metab* 2009;**94**:4961–4970.

Amiri M, Ramezani Tehrani F, Behboudi-Gandevani S, Bidhendi-Yarandi R, Carmina E. Risk of hypertension in women with polycystic ovary syndrome: a systematic review, meta-analysis and meta-regression. *Reprod Biol Endocrinol* 2020;**18**:23.

Anagnostis P, Paparodis RD, Bosdou JK, Bothou C, Macut D, Goulis DG, Livadas S. Risk of type 2 diabetes mellitus in polycystic ovary syndrome is associated with obesity: a meta-analysis of observational studies. *Endocrine* 2021;**74**:245–253.

Anderson KE, Sellers TA, Chen PL, Rich SS, Hong CP, Folsom AR. Association of Stein-Leventhal syndrome with the incidence of postmenopausal breast carcinoma in a large prospective study of women in Iowa. *Cancer* 1997;**79**:494–499.

Armeni E, Stamatelopoulos K, Rizos D, Georgiopoulos G, Kazani M, Kazani A, Kolyviras A, Stellos K, Panoulis K, Alexandrou A *et al.* Arterial stiffness is increased in asymptomatic nondiabetic postmenopausal women with a polycystic ovary syndrome phenotype. *J Hypertens* 2013;**31**:1998–2004.

Asuncion M, Calvo RM, San Millan JL, Sancho J, Avila S, Escobar-Morreale HF. A prospective study of the prevalence of the polycystic ovary syndrome in unselected Caucasian women from Spain. *J Clin Endocrinol Metab* 2000;**85**:2434–2438.

Azziz R, Carmina E, Dewailly D, Diamanti-Kandarakis E, Escobar-Morreale HF, Futterweit W, Janssen OE, Legro RS, Norman RJ, Taylor AE *et al.*; Androgen Excess Society. Position statement: criteria for defining polycystic ovary syndrome as a predominantly hyperandrogenic syndrome: an Androgen Excess Society guideline. *J Clin Endocrinol Metab* 2006;**91**:4237–4245.

Balen A. Polycystic ovary syndrome and cancer. *Hum Reprod Update* 2001;**7**:522–525.

Barry JA, Azizia MM, Hardiman PJ. Risk of endometrial, ovarian and breast cancer in women with polycystic ovary syndrome: a systematic review and meta-analysis. *Hum Reprod Update* 2014;**20**:748–758.

Birdsall MA, Farquhar CM. Polycystic ovaries in pre and post-menopausal women. *Clin Endocrinol (Oxf)* 1996;**44**:269–276.

Chan WP, Ngo DT, Sverdlov AL, Rajendran S, Stafford I, Heresztyn T, Chirkov YY, Horowitz JD. Premature aging of cardiovascular/platelet function in polycystic ovarian syndrome. *Am J Med* 2013;**126**:640.e1–7.

Cheang KI, Nestler JE, Futterweit W. Risk of cardiovascular events in mothers of women with polycystic ovary syndrome. *Endocr Pract* 2008;**14**:1084–1094.

Cibula D, Cifkova R, Fanta M, Poledne R, Zivny J, Skibova J. Increased risk of non-insulin dependent diabetes mellitus, arterial hypertension and coronary artery disease in perimenopausal women with a history of the polycystic ovary syndrome. *Hum Reprod* 2000;**15**:785–789.

Dahlgren E, Janson PO, Johansson S, Lapidus L, Lindstedt G, Tengborn L. Hemostatic and metabolic variables in women with polycystic ovary syndrome. *Fertil Steril* 1994;**61**:455–460.

Dahlgren E, Johansson S, Lindstedt G, Knutsson F, Oden A, Janson PO, Mattson LA, Crona N, Lundberg PA. Women with polycystic ovary syndrome wedge resected in 1956 to 1965: a long-term follow-up focusing on natural history and circulating hormones. *Fertil Steril* 1992;**57**:505–513.

Deniz A, Kehribar L. The association between knee osteoarthritis and polycystic ovary syndrome in postmenopausal women: preliminary results. *Eur Rev Med Pharmacol Sci* 2021;**25**:4895–4899.

Diamanti-Kandarakis E, Dunaif A. Insulin resistance and the polycystic ovary syndrome revisited: an update on mechanisms and implications. *Endocr Rev* 2012;**33**:981–1030.

Ding DC, Chen W, Wang JH, Lin SZ. Association between polycystic ovarian syndrome and endometrial, ovarian, and breast cancer: a population-based cohort study in Taiwan. *Medicine (Baltimore)* 2018;**97**:e12608.

Dokras A. Heart health in polycystic ovary syndrome: time to act on the data. *Fertil Steril* 2022;**117**:885–886.

Echiburu B, Crisosto N, Maliqueo M, Perez-Bravo F, de Guevara AL, Hernandez P, Cavada G, Rivas C, Clavel A, Sir-Petermann T. Metabolic profile in women with polycystic ovary syndrome across adult life. *Metabolism* 2016;**65**:776–782.

Elhassan YS, Idkowiak J, Smith K, Asia M, Gleeson H, Webster R, Arlt W, O'Reilly MW. Causes, patterns, and severity of androgen excess in 1205 consecutively recruited women. *J Clin Endocrinol Metab* 2018;**103**:1214–1223.

Elting MW, Korsen TJ, Bezemer PD, Schoemaker J. Prevalence of diabetes mellitus, hypertension and cardiac complaints in a follow-up study of a Dutch PCOS population. *Hum Reprod* 2001;**16**:556–560.

Escobar-Morreale HF, San Millan JL. Abdominal adiposity and the polycystic ovary syndrome. *Trends Endocrinol Metab* 2007;**18**:266–272.

Escobar-Morreale HF, Sanchon R, San Millan JL. A prospective study of the prevalence of nonclassical congenital adrenal hyperplasia among women presenting with hyperandrogenic symptoms and signs. *J Clin Endocrinol Metab* 2008;**93**:527–533.

Escobar-Morreale HF. Polycystic ovary syndrome: definition, aetiology, diagnosis and treatment. *Nat Rev Endocrinol* 2018;**14**:270–284.

Falcetta P, Benelli E, Molinaro A, Di Cosmo C, Bagattini B, Del Ghianda S, Salvetti G, Fiore E, Pucci E, Fruzzetti F *et al.* Effect of aging on clinical features and metabolic complications of women with polycystic ovary syndrome. *J Endocrinol Invest* 2021;**44**:2725–2733.

Forslund M, Landin-Wilhelmsen K, Brannstrom M, Dahlgren E. No difference in morbidity between perimenopausal women with PCOS with and without previous wedge resection. *Eur J Obstet Gynecol Reprod Biol* 2023;**285**:74–78.

Forslund M, Landin-Wilhelmsen K, Krantz E, Trimpou P, Schmidt J, Brännström M, Dahlgren E. Health-related quality of life in perimenopausal women with PCOS. *Clin Exp Obstet Gynecol* 2022a;**49**:52.

Forslund M, Landin-Wilhelmsen K, Schmidt J, Brannstrom M, Trimpou P, Dahlgren E. Higher menopausal age but no differences in parity in women with polycystic ovary syndrome compared with controls. *Acta Obstet Gynecol Scand* 2019;**98**:320–326.

Forslund M, Landin-Wilhelmsen K, Trimpou P, Schmidt J, Brannstrom M, Dahlgren E. Type 2 diabetes mellitus in women with polycystic ovary syndrome during a 24-year period: importance of obesity and abdominal fat distribution. *Hum Reprod Open* 2020;**2020**:hoz042.

Forslund M, Schmidt J, Brannstrom M, Landin-Wilhelmsen K, Dahlgren E. Reproductive hormones and anthropometry: a follow-up of PCOS and controls from perimenopause to older than 80 years. *J Clin Endocrinol Metab* 2021;**106**:421–430.

Forslund M, Schmidt J, Brännström M, Landin-Wilhelmsen K, Dahlgren E. Morbidity and mortality in PCOS: a prospective follow-up up to a mean age above 80 years. *Eur J Obstet Gynecol Reprod Biol* 2022b;**271**:195–203.

Gabrielli L, Aquino EM. Polycystic ovary syndrome in Salvador, Brazil: a prevalence study in primary healthcare. *Reprod Biol Endocrinol* 2012;**10**:96.

Gabrielli L, de Almeida MCC, Aquino EM. Proposed criteria for the identification of polycystic ovary syndrome following menopause: an ancillary study of the Brazilian Longitudinal Study of Adult Health (ELSA-Brasil). *Maturitas* 2015;**81**:398–405.

Gammon MD, Thompson WD. Polycystic ovaries and the risk of breast cancer. *Am J Epidemiol* 1991;**134**:818–824.

Guoqing Z, Fang S, Lihui D, Bing Y, Qiaoling P, Yingting W, Jinxia L. Cerebral white matter lesions and silent cerebral infarcts in postmenopausal women with polycystic ovary syndrome. *Gynecol Endocrinol* 2016;**32**:655–658.

Helvaci N, Yildiz BO. The impact of ageing and menopause in women with polycystic ovary syndrome. *Clin Endocrinol (Oxf)* 2022;**97**:371–382.

Hendriks ML, Ket JC, Hompes PG, Homburg R, Lambalk CB. Why does ovarian surgery in PCOS help? Insight into the endocrine implications of ovarian surgery for ovulation induction in polycystic ovary syndrome. *Hum Reprod Update* 2007;**13**:249–264.

Hernández-Mijares A, Bañuls C, Gómez-Balaguer M, Bergoglio M, Victor VM, Rocha M. Influence of obesity on atherogenic dyslipidemia in women with polycystic ovary syndrome. *Eur J Clin Invest* 2013;**43**:549–556.

Iatrakis G, Zervoudis S, Saviolakis A, Troulos M, Antoniou E, Sarantaki A, Lykeridou K, Kourounis G. Women younger than 50 years with endometrial cancer. *Eur J Gynaecol Oncol* 2006;**27**:399–400.

Karjula S, Morin-Papunen L, Auvinen J, Ruokonen A, Puukka K, Franks S, Jarvelin MR, Tapanainen JS, Jokelainen J, Miettunen J *et al.* Psychological distress is more prevalent in fertile age and premenopausal women with PCOS symptoms: 15-year follow-up. *J Clin Endocrinol Metab* 2017;**102**:1861–1869.

Karjula S, Morin-Papunen L, Franks S, Auvinen J, Jarvelin MR, Tapanainen JS, Jokelainen J, Miettunen J, Piltonen TT. Population-based data at ages 31 and 46 show decreased HRQoL and life satisfaction in women with PCOS symptoms. *J Clin Endocrinol Metab* 2020;**105**:1814–1826.

Kim C, Aroda VR, Goldberg RB, Younes N, Edelstein SL, Carrion-Petersen M, Ehrmann DA; Diabetes Prevention Program Outcomes Study Group. Androgens, irregular menses, and risk of diabetes and coronary artery calcification in the Diabetes Prevention Program. *J Clin Endocrinol Metab* 2018;**103**:486–496.

Kim J, Mersereau JE, Khankari N, Bradshaw PT, McCullough LE, Cleveland R, Shantakumar S, Teitelbuam SL, Neugut AI, Senie RT *et al.* Polycystic ovarian syndrome (PCOS), related symptoms/sequelae, and breast cancer risk in a population-based case-control study. *Cancer Causes Control* 2016;**27**:403–414.

Krentz AJ, Barrett-Connor E. Reduced serum ghrelin in a putative postmenopausal polycystic ovary syndrome phenotype. *Fertil Steril* 2009;**92**:1753–1754.

Krentz AJ, von Muhlen D, Barrett-Connor E. Searching for polycystic ovary syndrome in postmenopausal women: evidence of a dose-effect association with prevalent cardiovascular disease. *Menopause* 2007;**14**:284–292.

Krentz AJ, von Muhlen D, Barrett-Connor E. Adipocytokine profiles in a putative novel postmenopausal polycystic ovary syndrome (PCOS) phenotype parallel those in premenopausal PCOS: the Rancho Bernardo Study. *Metabolism* 2012;**61**:1238–1241.

Kujanpää L, Arffman RK, Pesonen P, Korhonen E, Karjula S, Järvelin MR, Franks S, Tapanainen JS, Morin-Papunen L, Piltonen TT. Women with polycystic ovary syndrome are burdened with multimorbidity and medication use independent of body mass index at late fertile age: a population-based cohort study. *Acta Obstet Gynecol Scand* 2022;**101**:728–736.

Legro RS. Polycystic ovary syndrome and cardiovascular disease: a premature association? *Endocr Rev* 2003;**24**:302–312.

Li J, Eriksson M, Czene K, Hall P, Rodriguez-Wallberg KA. Common diseases as determinants of menopausal age. *Hum Reprod* 2016;**31**:2856–2864.

Livadas S, Macut D, Bothou C, Kuliczkowska-Płaksej J, Vryonidou A, Bjekic-Macut J, Mouslech Z, Milewicz A, Panidis D. Insulin resistance, androgens, and lipids are gradually improved in an age-dependent manner in lean women with polycystic ovary syndrome: insights from a large Caucasian cohort. *Hormones (Athens)* 2020;**19**:531–539.

Luque-Ramirez M, Alpanes M, Sanchon R, Fernandez-Duran E, Ortiz-Flores AE, Escobar-Morreale HF. Referral bias in female functional hyperandrogenism and polycystic ovary syndrome. *Eur J Endocrinol* 2015;**173**:603–610.

Luque-Ramirez M, Alvarez-Blasco F, Mendieta-Azcona C, Botella-Carretero JI, Escobar-Morreale HF. Obesity is the major determinant of the abnormalities in blood pressure found in young women with the polycystic ovary syndrome. *J Clin Endocrinol Metab* 2007;**92**:2141–2148.

Luque-Ramirez M, Marti D, Fernandez-Duran E, Alpanes M, Alvarez-Blasco F, Escobar-Morreale HF. Office blood pressure, ambulatory blood pressure monitoring, and echocardiographic abnormalities in women with polycystic ovary syndrome: role of obesity and androgen excess. *Hypertension* 2014;**63**:624–629.

Margolin E, Zhornitzki T, Kopernik G, Kogan S, Schattner A, Knobler H. Polycystic ovary syndrome in post-menopausal women–marker of the metabolic syndrome. *Maturitas* 2005;**50**:331–336.

Markopoulos MC, Kassi E, Alexandraki KI, Mastorakos G, Kaltsas G. Hyperandrogenism after menopause. *Eur J Endocrinol* 2015;**172**: R79–R91.

Markopoulos MC, Rizos D, Valsamakis G, Deligeoroglou E, Grigoriou O, Chrousos GP, Creatsas G, Mastorakos G. Hyperandrogenism in women with polycystic ovary syndrome persists after menopause. *J Clin Endocrinol Metab* 2011;**96**:623–631.

Markopoulos MC, Valsamakis G, Kouskouni E, Boutsiadis A, Papassotiriou I, Creatsas G, Mastorakos G. Study of carbohydrate metabolism indices and adipocytokine profile and their relationship with androgens in polycystic ovary syndrome after menopause. *Eur J Endocrinol* 2013;**168**:83–90.

Matsuda M, DeFronzo RA. Insulin sensitivity indices obtained from oral glucose tolerance testing: comparison with the euglycemic insulin clamp. *Diabetes Care* 1999;**22**:1462–1470.

McGuinness LA, Higgins JPT. Risk-of-bias VISualization (robvis): an R package and Shiny web app for visualizing risk-of-bias assessments. *Res Synth Methods* 2021;**12**:55–61.

McKinlay SM, Brambilla DJ, Posner JG. The normal menopause transition. *Maturitas* 1992;**14**:103–115.

Merz CNB, Shaw LJ, Azziz R, Stanczyk FZ, Sopko G, Braunstein GD, Kelsey SF, Kip KE, Cooper-DeHoff RM, Johnson BD *et al.* Cardiovascular disease and 10-year mortality in postmenopausal women with clinical features of polycystic ovary syndrome. *J Womens Health (Larchmt)* 2016;**25**:875–881.

Meun C, Franco OH, Dhana K, Jaspers L, Muka T, Louwers Y, Ikram MA, Fauser B, Kavousi M, Laven JSE. High androgens in postmenopausal women and the risk for atherosclerosis and cardiovascular disease: the Rotterdam Study. *J Clin Endocrinol Metab* 2018;**103**:1622–1630.

Meun C, Gunning MN, Louwers YV, Peters H, Roos-Hesselink J, Roeters van Lennep J, Rueda Ochoa OL, Appelman Y, Lambalk N, Boersma E *et al.*; CREW Consortium. The cardiovascular risk profile of middle-aged women with polycystic ovary syndrome. *Clin Endocrinol (Oxf)* 2020;**92**:150–158.

Minooee S, Ramezani Tehrani F, Rahmati M, Mansournia MA, Azizi F. Prediction of age at menopause in women with polycystic ovary syndrome. *Climacteric* 2018;**21**:29–34.

Mulhall S, Andel R, Anstey KJ. Variation in symptoms of depression and anxiety in midlife women by menopausal status. *Maturitas* 2018;**108**:7–12.

Naderpoor N, Shorakae S, de Courten B, Misso ML, Moran LJ, Teede HJ. Metformin and lifestyle modification in polycystic ovary syndrome: systematic review and meta-analysis. *Hum Reprod Update* 2016;**22**:408–409.

Ollila MM, Kaikkonen K, Järvelin MR, Huikuri HV, Tapanainen JS, Franks S, Piltonen TT, Morin-Papunen L. Self-reported polycystic ovary syndrome is associated with hypertension: a Northern Finland Birth Cohort 1966 Study. *J Clin Endocrinol Metab* 2019;**104**: 1221–1231.

Ollila MM, Piltonen T, Puukka K, Ruokonen A, Järvelin MR, Tapanainen JS, Franks S, Morin-Papunen L. Weight gain and dyslipidemia in early adulthood associate with polycystic ovary syndrome: prospective cohort study. *J Clin Endocrinol Metab* 2016;**101**: 739–747.

Ollila MM, West S, Keinänen-Kiukaanniemi S, Jokelainen J, Auvinen J, Puukka K, Ruokonen A, Järvelin MR, Tapanainen JS, Franks S *et al.* Overweight and obese but not normal weight women with PCOS are at increased risk of Type 2 diabetes mellitus – a prospective, population-based cohort study. *Hum Reprod* 2017;**32**: 423–431.

Ortiz-Flores AE, Luque-Ramirez M, Fernandez-Duran E, Alvarez-Blasco F, Escobar-Morreale HF. Diagnosis of disorders of glucose tolerance in women with polycystic ovary syndrome (PCOS) at a tertiary care center: fasting plasma glucose or oral glucose tolerance test? *Metabolism* 2019;**93**:86–92.

Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, Shamseer L, Tetzlaff JM, Akl EA, Brennan SE *et al.* The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 2021;**372**:n71.

Parazzini F, La Vecchia C, Franceschi S, Talamini R, Negri E, Crosignani PG. Association of Stein-Leventhal syndrome with the incidence of postmenopausal breast carcinoma in a large prospective study of women in Iowa. *Cancer* 1997;**80**:1360–1362.

Persson S, Ubhayasekera KA, Bergquist J, Turkmen S, Poromaa IS, Elenis E. Hyperandrogenic symptoms are a persistent suffering in midlife women with PCOS; a prospective cohort study in Sweden. *Biomedicines* 2023;**11**:96.

Pinola P, Piltonen TT, Puurunen J, Vanky E, Sundstrom-Poromaa I, Stener-Victorin E, Ruokonen A, Puukka K, Tapanainen JS, Morin-Papunen LC. Androgen profile through life in women with polycystic ovary syndrome: a Nordic Multicenter Collaboration Study. *J Clin Endocrinol Metab* 2015;**100**:3400–3407.

Polonen J, Pinola P, Ronkainen J, Blakemore AI, Buxton JL, Tapanainen JS, Franks S, Piltonen TT, Sebert S, Morin-Papunen L. Polycystic ovary syndrome and leukocyte telomere length: cross-sectional and longitudinal changes. *Eur J Endocrinol* 2022;**187**:651–661.

Polotsky AJ, Allshouse A, Crawford SL, Harlow SD, Khalil N, Santoro N, Legro RS. Relative contributions of oligomenorrhea and hyperandrogenemia to the risk of metabolic syndrome in midlife women. *J Clin Endocrinol Metab* 2012;**97**:E868–E877.

Polotsky AJ, Allshouse AA, Crawford SL, Harlow SD, Khalil N, Kazlauskaite R, Santoro N, Legro RS. Hyperandrogenic oligomenorrhea and metabolic risks across menopausal transition. *J Clin Endocrinol Metab* 2014;**99**:2120–2127.

Puurunen J, Piltonen T, Jaakkola P, Ruokonen A, Morin-Papunen L, Tapanainen JS. Adrenal androgen production capacity remains high up to menopause in women with polycystic ovary syndrome. *J Clin Endocrinol Metab* 2009;**94**:1973–1978.

Puurunen J, Piltonen T, Morin-Papunen L, Perheentupa A, Jarvela I, Ruokonen A, Tapanainen JS. Unfavorable hormonal, metabolic, and inflammatory alterations persist after menopause in women with PCOS. *J Clin Endocrinol Metab* 2011;**96**:1827–1834.

Ramezani Tehrani F, Amiri M, Behboudi-Gandevani S, Bidhendi-Yarandi R, Carmina E. Cardiovascular events among reproductive and menopausal age women with polycystic ovary syndrome: a systematic review and meta-analysis. *Gynecol Endocrinol* 2020;**36**:12–23.

Schmidt J, Brannstrom M, Landin-Wilhelmsen K, Dahlgren E. Reproductive hormone levels and anthropometry in postmenopausal women with polycystic ovary syndrome (PCOS): a 21-year follow-up study of women diagnosed with PCOS around 50 years ago and their age-matched controls. *J Clin Endocrinol Metab* 2011a; **96**:2178–2185.

Schmidt J, Dahlgren E, Brannstrom M, Landin-Wilhelmsen K. Body composition, bone mineral density and fractures in late postmenopausal women with polycystic ovary syndrome – a long-term follow-up study. *Clin Endocrinol (Oxf)* 2012;**77**:207–214.

Schmidt J, Dahlgren E, Bryman I, Berntorp K, Trimpou P, Wilhelmsen L, Landin-Wilhelmsen K. High androgen levels protect against hypothyroidism. *Acta Obstet Gynecol Scand* 2017;**96**:39–46.

Schmidt J, Landin-Wilhelmsen K, Brannstrom M, Dahlgren E. Cardiovascular disease and risk factors in PCOS women of post-menopausal age: a 21-year controlled follow-up study. *J Clin Endocrinol Metab* 2011b;**96**:3794–3803.

Snyder ML, Shields KJ, Korytkowski MT, Sutton-Tyrrell K, Talbott EO. Complement protein C3 and coronary artery calcium in middle-aged women with polycystic ovary syndrome and controls. *Gynecol Endocrinol* 2014;**30**:511–515.

Sterne JAC, Becker BJ, Egger M. The funnel plot. In: Rothstein HR, Sutton AJ, Borenstein M (eds). *Publication Bias in Meta-Analysis – Prevention, Assessment and Adjustments*. Chichester, UK: Willey & Sons, 2005.

Talbott E, Clerici A, Berga SL, Kuller L, Guzick D, Detre K, Daniels T, Engberg RA. Adverse lipid and coronary heart disease risk profiles in young women with polycystic ovary syndrome: results of a case-control study. *J Clin Epidemiol* 1998;**51**:415–422.

Talbott EO, Guzick DS, Sutton-Tyrrell K, McHugh-Pemu KP, Zborowski JV, Remsberg KE, Kuller LH. Evidence for association between polycystic ovary syndrome and premature carotid atherosclerosis in middle-aged women. *Arterioscler Thromb Vasc Biol* 2000;**20**:2414–2421.

Talbott EO, Zborowski J, Rager J, Stragand JR. Is there an independent effect of polycystic ovary syndrome (PCOS) and menopause on the prevalence of subclinical atherosclerosis in middle aged women? *Vasc Health Risk Manag* 2008;**4**:453–462.

Talbott EO, Zborowski JV, Boudreaux MY, McHugh-Pemu KP, Sutton-Tyrrell K, Guzick DS. The relationship between C-reactive protein and carotid intima-media wall thickness in middle-aged women with polycystic ovary syndrome. *J Clin Endocrinol Metab* 2004a;**89**:6061–6067.

Talbott EO, Zborowski JV, Rager JR, Boudreaux MY, Edmundowicz DA, Guzick DS. Evidence for an association between metabolic cardiovascular syndrome and coronary and aortic calcification among women with polycystic ovary syndrome. *J Clin Endocrinol Metab* 2004b;**89**:5454–5461.

Teede HJ, Misso ML, Costello MF, Dokras A, Laven J, Moran L, Piltonen T, Norman RJ; International PCOS Network. Recommendations from the international evidence-based guideline for the assessment and management of polycystic ovary syndrome. *Hum Reprod* 2018;**33**:1602–1618.

The Rotterdam ESHRE/ASRM-Sponsored PCOS Consensus Workshop Group. Revised 2003 consensus on diagnostic criteria and long-term health risks related to polycystic ovary syndrome (PCOS). *Hum Reprod* 2004;**19**:41–47.

Wan X, Wang W, Liu J, Tong T. Estimating the sample mean and standard deviation from the sample size, median, range and/or interquartile range. *BMC Med Res Methodol* 2014;**14**:135.

Wekker V, van Dammen L, Koning A, Heida KY, Painter RC, Limpens J, Laven JSE, Roeters van Lennep JE, Roseboom TJ, Hoek A. Long-term cardiometabolic disease risk in women with PCOS: a systematic review and meta-analysis. *Hum Reprod Update* 2020;**26**:942–960.

Welt CK, Carmina E. Lifecycle of polycystic ovary syndrome (PCOS): from in utero to menopause. *J Clin Endocrinol Metab* 2013;**98**:4629–4638.

Wickenheisser JK, Nelson-DeGrave VL, McAllister JM. Human ovarian theca cells in culture. *Trends Endocrinol Metab* 2006;**17**:65–71.

Wild RA, Carmina E, Diamanti-Kandarakis E, Dokras A, Escobar-Morreale HF, Futterweit W, Lobo R, Norman RJ, Talbott E, Dumesic DA. Assessment of cardiovascular risk and prevention of cardiovascular disease in women with the polycystic ovary syndrome: a consensus statement by the Androgen Excess and Polycystic Ovary Syndrome (AE-PCOS) Society. *J Clin Endocrinol Metab* 2010;**95**:2038–2049.

Wild RA, Rizzo M, Clifton S, Carmina E. Lipid levels in polycystic ovary syndrome: systematic review and meta-analysis. *Fertil Steril* 2011;**95**:1073–1079.e1–11.

Wild S, Pierpoint T, Jacobs H, McKeigue P. Long-term consequences of polycystic ovary syndrome: results of a 31 year follow-up study. *Hum Fertil (Camb)* 2000;**3**:101–105.

Winters SJ, Talbott E, Guzick DS, Zborowski J, McHugh KP. Serum testosterone levels decrease in middle age in women with the polycystic ovary syndrome. *Fertil Steril* 2000;**73**:724–729.

Yildiz BO, Bozdag G, Yapici Z, Esinler I, Yarali H. Prevalence, phenotype and cardiometabolic risk of polycystic ovary syndrome under different diagnostic criteria. *Hum Reprod* 2012;**27**:3067–3073.

Yin O, Zacur HA, Flaws JA, Christianson MS. Association between polycystic ovary syndrome and hot flash presentation during the midlife period. *Menopause* 2018;**25**:691–696.

Zawadzki JK, Dunaif A. Diagnostic criteria for polycystic ovary syndrome: towards a rational approach. In: Dunaif A, Givens JR, Haseltine FP, Merriam GR (eds). *Polycystic Ovary Syndrome*. Boston: Blackwell Scientific Publications, 1992.

Zhu S, Zhang B, Jiang X, Li Z, Zhao S, Cui L, Chen ZJ. Metabolic disturbances in non-obese women with polycystic ovary syndrome: a systematic review and meta-analysis. *Fertil Steril* 2019;**111**:168–177.

Zucchetto A, Serraino D, Polesel J, Negri E, De Paoli A, Dal Maso L, Montella M, La Vecchia C, Franceschi S, Talamini R. Hormone-related factors and gynecological conditions in relation to endometrial cancer risk. *Eur J Cancer Prev* 2009;**18**:316–321.